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Digital Twin for railway network simulation

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Chapter 1

Introduction

Railway networks are essential infrastructures for passenger mobility, freight transport, and sustainable transportation. In dense networks such as the Belgian railway system, operating trains efficiently and reliably is a complex task. Many trains, stations, tracks, and operational constraints interact continuously, while disruptions in one part of the network can propagate to other services and affect punctuality at a wider scale.

This complexity is both infrastructural and operational. Trains must follow predefined routes and timetables, share limited track capacity, respect safety constraints, and interact with elements such as switches, platforms, and stations. As a result, railway operators and researchers increasingly rely on digital tools to better understand, simulate, and analyse railway systems.

In this context, the concept of a digital twin has gained increasing attention. Informally, a digital twin can be understood as a virtual counterpart of a physical system, connected to data and used to observe, simulate, and analyse its behaviour. Applied to railways, such a representation may combine infrastructure data, train movements, operational constraints, and traffic conditions. Simulation is therefore a central component of this vision, since it provides a controlled environment in which railway operations can be reproduced, tested, and compared with real data.

This thesis focuses on the construction of a simulation-oriented railway digital twin using SUMO. The objective is not to build a complete real-time digital twin, but to study how heterogeneous railway infrastructure and operational datasets can be transformed into simulation-ready artefacts. The work also investigates how different levels of railway network representation influence the feasibility and realism of the resulting simulation.

1.1 Research Problem and Objectives

Despite the availability of railway data and simulation platforms, constructing a realistic railway simulation from real-world datasets remains difficult. Railway data is heterogeneous by nature: infrastructure datasets may describe stations, tracks, switches, platforms, and geographical geometries, while operational datasets describe train movements, schedules, arrival and departure times, and punctuality records. These datasets do not always share the same structure, granularity, or level of detail. Integrating them into a single simulation framework therefore requires several processing, modelling, and approximation steps.

A first challenge concerns the representation of the railway infrastructure. At a high level, the network can be represented as a graph in which stations are nodes and railway links are edges. This abstraction is useful because it simplifies route reconstruction and supports large-scale simulation. However, it does not represent detailed infrastructure elements such as platforms, switches, parallel tracks, or station-level routing constraints.

A second challenge appears when moving toward a more detailed representation. At the micro level, tracks, platforms, switches, and routing structures must be represented explicitly. This provides a more realistic description of the infrastructure, but it also requires a topologically consistent network. Platforms must be correctly associated with tracks, and valid paths must exist between consecutive stations. Even small inconsistencies in the reconstructed infrastructure can prevent trains from being routed correctly.

A third challenge concerns the integration of real operational data. Punctuality data provides information about train stops, planned and observed times, and delays, but it does not directly provide the exact physical route followed by each train. Train journeys must therefore be reconstructed from station sequences and mapped onto the generated simulation network. This process is relatively robust at the macro level, but becomes more fragile at the micro level because each movement must correspond to a valid sequence of physical track segments.

These challenges are addressed through the following research sub-questions:

RQ1. How can heterogeneous railway infrastructure and operational datasets be transformed into SUMO-compatible simulation components?

RQ2. To what extent can a macro-level station-to-station representation support the integration of real operational data into railway simulation?

RQ3. To what extent can a micro-level infrastructure representation support detailed railway simulation based on tracks, platforms, switches, and routing structures?

RQ4. What are the main limitations and failure points encountered when integrating real operational data into micro-level railway simulation?

The main objective of this thesis is to design and implement a data-driven pipeline for constructing a simulation-oriented railway digital twin. This pipeline is evaluated through two complementary modelling approaches. The macro-level model represents the railway network as a station-to-station graph and is used to validate the integration of real operational data into simulation. The micro-level model represents the infrastructure in greater detail and is used to study the feasibility of more realistic railway simulation.

The expected outcome is therefore not a perfect reproduction of the Belgian railway network, but a structured and evaluated prototype showing how railway data can be transformed into SUMO-compatible simulation models. The contribution lies in the construction of the modelling pipeline, the comparison between macro-level and micro-level representations, and the identification of the technical challenges that must be addressed to progress toward more complete railway digital twins.

1.2 Thesis Structure

The remainder of this thesis is organized as follows.

Chapter 2 presents the state of the art. It introduces the concept of digital twins, first from a general perspective and then in the context of railway systems. It also reviews the main components of railway infrastructure and operations, discusses common approaches for railway network representation, and presents existing railway simulation tools, with particular attention to SUMO.

Chapter 3 describes the proposed methodology. It positions the work with respect to the research gaps identified in the literature, presents the overall framework of the approach, justifies the choice of SUMO, and describes the infrastructure and operational datasets used in the implementation.

Chapter 4 details the implementation of the proposed approach. It is divided into two main parts. The first presents the macro-level model, including data processing, network generation, trip generation, and simulation. The second presents the micro-level model, including infrastructure modelling, switch detection, platform detection, track segmentation, dual graph construction, trip generation, and simulation.

Chapter 5 presents the results obtained from the implemented models. It analyses dataset coverage, infrastructure modelling quality, simulation results, and computational performance. Particular attention is given to the difference between successful macro-level integration and the difficulties encountered when using real operational data at the micro level.

Chapter 6 discusses the main limitations of the proposed methodology. It provides a more detailed analysis of issues related to data quality, modelling assumptions, route reconstruction, SUMO constraints, and scalability.

Chapter 7 concludes the thesis by summarizing the main contributions and findings. It also discusses the practical implications of the work and presents future research directions, including counterfactual simulation, track work planning, real-time forecasting, incident propagation, and framework generalization.

Chapter 2

State of the Art

2.1 Digital Twins

Digital twins have become a central concept in the digitalization of complex industrial systems [19]. The term is used in many domains, including manufacturing, aerospace, healthcare, smart cities, energy systems, and transportation [19, 35, 33]. Although definitions vary depending on the application domain, the general principle remains the same: a digital twin is a digital representation of a physical system connected to data and used to analyse, simulate, monitor, or support decisions about the real system [24, 19].

In the context of this thesis, the concept is particularly relevant because railway networks are complex, dynamic, and data-rich systems. They combine physical infrastructure, moving vehicles, operational rules, timetables, delays, maintenance constraints, and safety requirements. A digital twin can therefore provide a structured framework to connect infrastructure data, operational data, simulation models, and analytical tools into a coherent representation of the railway system.

2.1.1 Definition, Capabilities and Limitations

The concept of digital twin was first introduced in the context of product lifecycle management and later extended to many engineering and industrial domains. A common definition describes a digital twin as a virtual representation of a physical object, process, or system that is connected to its physical counterpart through data exchange and can be used throughout the lifecycle of the system. This definition emphasizes that a digital twin is not only a static model or a visualization tool. Its distinctive feature is the relationship between the physical system and its digital counterpart.

A digital twin can be understood through three main components: the physical system, the virtual system, and the connection between them. The physical system corresponds to the real-world object or process being represented. The virtual system describes its structure, behaviour, and state. The connection corresponds to the data flow linking both entities. This connection may be unidirectional, when the physical system only updates the virtual model, or bidirectional, when the digital model can also support decisions or actions applied to the physical system [34].

A more formal representation defines a digital twin as a system composed of several interacting elements [41, 19]:

$$DT = (P, V, D, C, S) \tag{2.1}$$

where P denotes the physical system, V the virtual representation, D the data associated

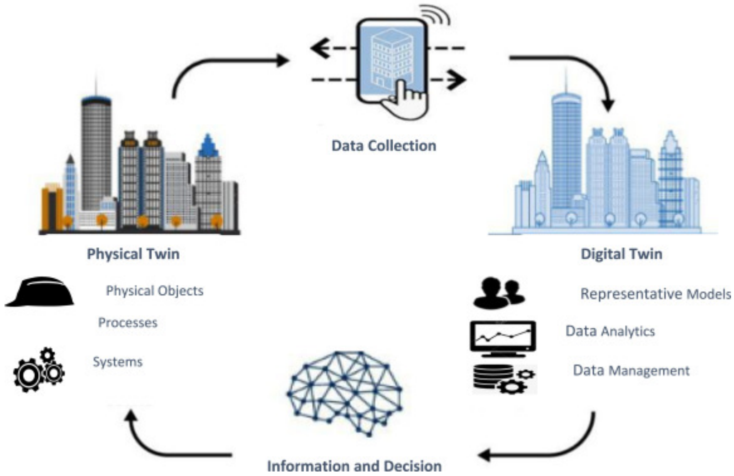

Figure 2.1: General architecture of a digital twin.

with the system, C the connection mechanisms between the physical and virtual entities, and S the services built on top of the twin, such as simulation, prediction, visualization, diagnosis, or optimization. This representation shows that a digital twin is not only a model, but an architecture combining data, models, and services to support decision-making.

It is also important to distinguish digital twins from related concepts. A digital model is a digital representation of a system without automatic data exchange with the physical system. A digital shadow receives data from the physical system, but the information flow remains mainly one-way. A digital twin represents a more advanced form, where the digital representation is regularly or continuously updated from the physical system and can support feedback, analysis, or operational decisions [30]. In this thesis, the proposed work is closer to a simulation-oriented digital twin foundation: it focuses on constructing coherent virtual railway representations from real infrastructure and operational data, which is a necessary step before a fully real-time and bidirectional railway digital twin can be achieved.

Digital twins are usually associated with several capabilities. Monitoring consists of observing the state of a system using sensor data, logs, operational records, or information systems. Simulation uses the digital representation to reproduce the behaviour of the physical system under controlled conditions [19, 39]. Prediction uses historical or real-time data to estimate future states, failures, delays, or performance indicators [19, 35, 33]. Optimization evaluates alternative scenarios in order to support better decisions. More advanced digital twins may also include control or actuation, where decisions derived from the digital model influence the physical system.

These capabilities make digital twins attractive for complex systems, but they also introduce limitations. A digital twin is only as reliable as the data, models, assumptions, and synchronization mechanisms on which it is based. Incomplete, outdated, noisy, or inconsistent data may produce misleading results. Similarly, a model that is too simplified may ignore important physical or operational constraints, while a model that is too detailed may become computationally expensive and difficult to maintain. Validation is also a major challenge, since a digital twin must be compared with real observations, even though real systems often include hidden variables, incomplete measurements, operational exceptions, and human decisions that are not fully captured in the available data. For this reason, digital twins are often developed incrementally, starting from static representation and simulation before integrating real-time data, predictive models, and optimization services.

2.1.2 Recent Developments and Applications

The development of digital twins has accelerated with the progress of several enabling technologies. The IoTs allows physical systems to generate continuous streams of data [19]. Cloud computing and distributed architectures provide the infrastructure required to store and process large volumes of information. Artificial intelligence and machine learning make it possible to extract patterns, detect anomalies, and predict future states. Simulation tools provide a controlled environment in which alternative scenarios can be tested without disrupting the real system. Finally, visualization technologies, such as dashboards, 3D interfaces, and geographic information systems, help users interpret the state and behaviour of the DT.

Digital twins were first widely adopted in manufacturing, where they support product design, production monitoring, maintenance planning, and process optimization. In smart manufacturing, a digital twin can represent a machine, a production line, or an entire factory. It can be used to monitor equipment status, simulate production scenarios, predict failures, and improve resource allocation [39]. This domain is important in the literature because it provides many conceptual foundations later reused in other fields.

In aerospace, mechanical engineering, and energy systems, digital twins are often used to monitor critical assets and support predictive maintenance. Aircraft, engines, turbines, power plants, and renewable energy infrastructure generate large amounts of operational data during their lifecycle. A digital twin can combine this data with physics-based or data-driven models in order to estimate wear, detect abnormal behaviour, and plan maintenance interventions before failures occur [33]. Digital twins are also increasingly studied in healthcare and smart cities, where they can represent medical devices, hospital workflows, patient-specific models, urban infrastructure, mobility systems, and environmental condition [35].

These capabilities make digital twins attractive for complex systems, but they also introduce limitations. A digital twin is only as reliable as the data, models, assumptions, and synchronization mechanisms on which it is based. Incomplete, outdated, noisy, or inconsistent data may produce misleading results. Similarly, a model that is too simplified may ignore important physical or operational constraints, while a model that is too detailed may become computationally expensive and difficult to maintain. Validation is also a major challenge, since a digital twin must be compared with real observations, even though real systems often include hidden variables, incomplete measurements, operational exceptions, and human decisions that are not fully captured in the available data. For this reason, digital twins are often developed incrementally, starting from static representation and simulation before integrating real-time data, predictive models, and optimization services.

2.1.3 Applications in Railway networks

Railway systems are particularly suitable candidates for digital twin approaches. They combine physical infrastructure, operational processes, moving assets, strict safety constraints, and large volumes of data. A railway network includes tracks, switches, signals, stations, platforms, rolling stock, timetables, passenger flows, maintenance operations, and traffic management rules. These elements interact continuously, and a local disturbance can propagate through the network. Railway operators therefore require tools able to represent the state of the network, simulate traffic behaviour, evaluate disruptions, and support planning decisions [44, 44].

Railway digital twins can be applied at different levels of granularity. At the asset level, a digital twin can represent individual components such as tracks, switches, bridges, tunnels, trains, or signalling equipment. This type of twin is mainly useful for condition monitoring and predictive maintenance, where sensor data, inspection records, and maintenance logs are used to estimate asset health and anticipate interventions. At the infrastructure level, a digital twin can represent the topology of the railway network, including tracks, stations, platforms, junctions, and operational segments. This representation is necessary to analyse accessibility, capacity, routing possibilities, and infrastructure conflicts. At the operational level, a digital twin can represent train movements, schedules, delays, and traffic interactions, making it possible to analyse circulation patterns, delay propagation, and the impact of infrastructure constraints on service quality.

Several railway applications can be identified. Predictive maintenance is one of the most common, since railway infrastructure is expensive to inspect and failures can strongly affect safety and reliability. Traffic management is another important application: by reproducing train movements in a virtual environment, a digital twin can help evaluate congestion, detect bottlenecks, and test operational strategies. DTs can also support delay analysis and propagation modelling, since a delay affecting one train may influence many others. Finally, simulation-based digital twins are useful for scenario testing, such as timetable changes, temporary track closures, infrastructure works, disruptions, or emergency response strategies.

However, railway digital twins also raise specific challenges. Railway systems are safety-critical, which means that simplified models must be used carefully. Accurate representation of tracks, switches, signals, platforms, and routing constraints becomes necessary when the objective is to reproduce detailed operational behaviour. Data availability is another major issue, since infrastructure data may be incomplete, private, heterogeneous, or difficult to align with operational data. In addition, railway systems operate at several spatial and temporal scales: long-distance network connectivity, station-level infrastructure, and dynamic train operations may all need to be represented within the same framework.

For these reasons, railway digital twins often require a layered approach. A macro-level representation can model station-to-station connectivity and global train movements, while a micro-level representation can model detailed infrastructure such as tracks, switches, and platforms. This distinction is directly reflected in the methodology of this thesis. The macro-level model provides a simplified and scalable representation of the railway network, whereas the micro-level model aims to capture more detailed infrastructure constraints. Together, these two levels contribute to the construction of a simulation-oriented foundation for a railway digital twin.

In summary, digital twins provide a relevant conceptual and technical framework for railway simulation. They offer a way to combine infrastructure data, operational data, simulation models, and analytical tools into a unified system. However, building a complete railway digital twin requires several intermediate steps: reconstructing the infrastructure, generating valid train routes, integrating real operational data, validating simulation outputs, and identifying the limitations of the model. This thesis focuses on these foundational steps, with the objective of preparing the ground for future applications such as real-time forecasting, disruption analysis, and decision support.

2.2 Railway Systems

Railway systems are complex socio-technical systems composed of physical infrastructure, rolling stock, operational rules, control systems, and data-driven decision processes. Unlike road traffic, where vehicles usually have a high degree of freedom in their trajectories, trains are constrained to follow predefined tracks, respect signalling rules, and operate according to planned timetables. This makes railway systems highly structured, but also highly constrained.

Understanding railway systems is essential for the construction of a railway digital twin. A digital twin does not only require a geometric or topological description of the network. It must also represent how trains move, how infrastructure is used, how conflicts are avoided, and how delays may propagate. For this reason, the following sections introduce the main infrastructure components, operational principles, network representations, and data sources required for railway modelling and simulation.

2.2.1 Railway infrastructure

Railway infrastructure refers to the physical and technical components that allow trains to circulate safely and efficiently. It includes tracks, switches, junctions, stations, platforms, signalling systems, electrification equipment, and communication systems. From a modelling perspective, infrastructure defines both the spatial structure of the network and the constraints under which train movements can occur.

The level of detail used to represent infrastructure depends on the objective of the model. A high-level model may only represent stations and the connections between them, which is sufficient for analysing large-scale connectivity or approximate train movements. A more detailed model must represent tracks, switches, platforms, signals, and route conflicts, which is necessary for studying station-level operations, capacity, and realistic routing constraints [3].

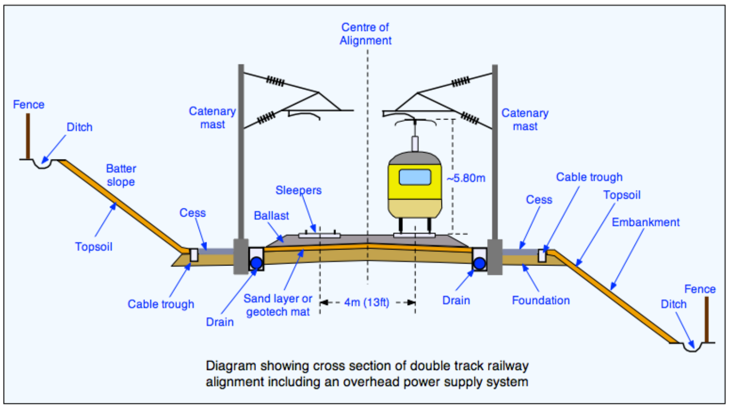

Figure 2.2: Schematic representation of railway infrastructure components.

Tracks

Tracks are the fundamental physical components of a railway network. They guide train movements and define the possible spatial paths through the infrastructure. From an operational point of view, a track is not only a physical object but also a capacity resource, since only a limited number of trains can use the same section within a given time interval [11].

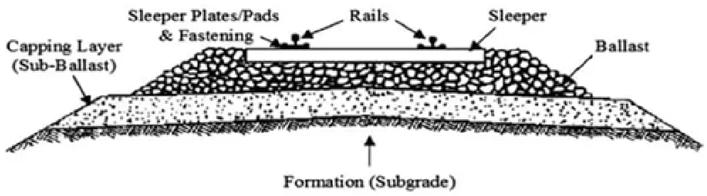

Figure 2.3: Railway track structure.

In modelling terms, tracks can be represented at different levels of abstraction. At a macroscopic level, a track may be abstracted as a connection between two stations. This is useful for station-to-station connectivity and large-scale train movement analysis. At a microscopic level, tracks must be represented as individual segments with geometry, direction, length, speed constraints, and connections to neighbouring elements. This level of detail is required when the objective is to compute feasible routes through complex station areas or to represent access to switches and platforms.

Track directionality is also important. Some railway lines are designed for traffic in one direction only, while others allow bidirectional operation. Even when tracks are physically bidirectional, signalling and operational rules may restrict their use. Therefore, a railway model must distinguish between physical connectivity and operationally usable connectivity.

Switches and Junctions

Switches, also called points or turnouts, allow trains to move from one track to another. They are essential for routing because they introduce branching possibilities in the network. Switches enable trains to enter stations, change tracks, access sidings, overtake other trains, or follow alternative routes [9, 10].

From a modelling perspective, a switch can be seen as a local decision point in the infrastructure graph. It connects an incoming track to several possible outgoing tracks depending on its configuration. Junctions are larger structures composed of several switches and crossings, often located near stations, yards, or intersections between railway lines.

Switches introduce important operational constraints. A train can only pass through a switch if it is correctly aligned for the intended route, and two trains cannot use conflicting routes through the same junction at the same time. These constraints are normally enforced by signalling and interlocking systems. In detailed railway simulations, switches and junctions must therefore be modelled not only as graph connections, but also as shared resources that may generate conflicts.

The correct identification of switches is particularly important for digital twin construction. Missing switches may prevent valid routes from being found, while incorrectly

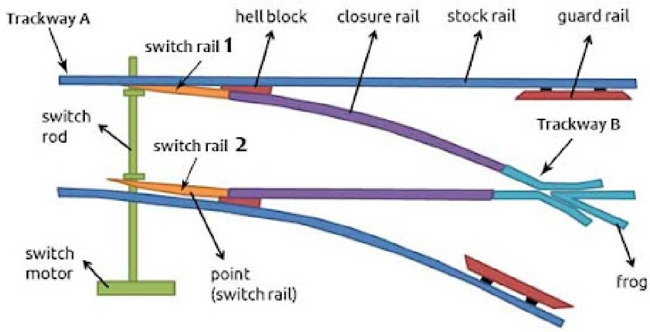

Figure 2.4: Railway switch structure.

detected switches may create unrealistic routing possibilities. This is one of the reasons why micro-level railway modelling is more difficult than macro-level modelling.

Stations and Platforms

Stations are operational nodes where trains may stop, passengers may board or alight, and services may be regulated. They are often among the most complex elements of a railway network because they combine tracks, platforms, switches, signals, passenger flows, and timetable constraints [20].

In a macroscopic representation, a station can be modelled as a single node. This is useful for analysing station-to-station connectivity or reconstructing train journeys from operational data. However, such a representation hides the internal structure of the station. It does not indicate which platform a train uses, whether two trains can stop simultaneously, or whether access routes to different platforms conflict.

In a microscopic representation, a station must be decomposed into tracks, platforms, stopping positions, and access routes. Platforms define where passenger exchanges occur and are important for simulation because train stops must be attached to specific tracks or lanes. If a platform is incorrectly associated with a track, the train may stop at the wrong location or may not be able to reach the stop at all.

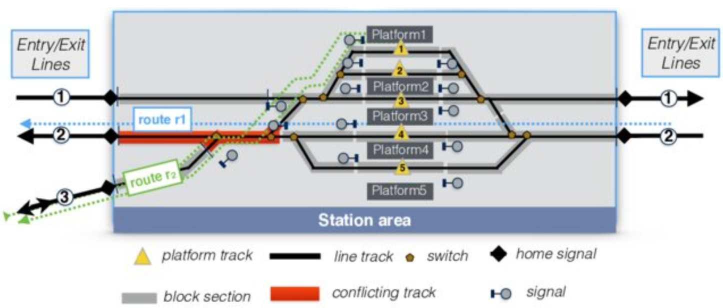

Figure 2.5: Example of a station layout

Stations also influence capacity. Dwell time, platform availability, route conflicts at station throats, and the number of available tracks determine how many trains can be handled within a given period. Large stations may become bottlenecks even if the lines around them have sufficient capacity. For this reason, station modelling is a key component of realistic railway simulation.

In this thesis, stations are used differently in the macro-level and micro-level approaches. In the macro-level model, stations are represented as nodes connected by edges. In the micro-level model, stations are associated with platforms and track segments. This distinction makes it possible to compare a simplified connectivity-based representation with a more detailed infrastructure-based representation.

Stations also influence capacity. Dwell time, platform availability, route conflicts at station throats, and the number of available tracks determine how many trains can be handled within a given period. Large stations may therefore become bottlenecks even when the surrounding lines have sufficient capacity. In this thesis, stations are represented differently depending on the modelling level: as single nodes in the macro-level model, and as platform-track associations in the micro-level model.

Signaling Systems

Signalling systems ensure the safe movement of trains through the railway network. Their role is to prevent conflicting train movements and maintain safe separation between trains. They communicate movement authorities, speed restrictions, route availability, and stopping instructions to train drivers or onboard systems [4].

Traditional signalling systems divide the railway line into block sections. Under normal operation, only one train is allowed to occupy a block section at a time. A train may proceed only if the section ahead is clear and if the route has been correctly set. This principle prevents rear-end collisions and conflicting movements. In more advanced systems, movement authority may be transmitted directly to the train through cab signalling or automatic train protection systems.

Traditional signalling systems divide the railway line into block sections. Under normal operation, only one train is allowed to occupy a block section at a time. A train may proceed only if the section ahead is clear and if the route has been correctly set. More advanced systems may transmit movement authority directly to the train through cab signalling or automatic train protection systems.

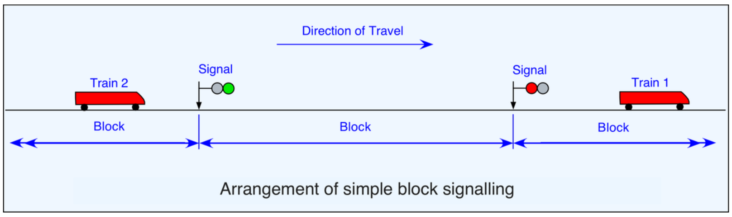

Figure 2.6: Simplified block signalling principle

In Europe, signalling and train control are strongly influenced by interoperability requirements [17]. The European Rail Traffic Management System (ERTMS) and the European Train Control System (ETCS) aim to standardize train control and signalling across countries. Such systems are relevant for digital twin research because they generate and rely on structured operational data and because they influence how train movements can be represented.

From a simulation perspective, signalling can be represented at different levels of detail. A simple simulation may ignore signalling and only check whether routes exist. A more realistic simulation must represent block occupation, signal states, route reservations, interlocking constraints, and train separation. The level of signalling detail has a direct impact on the realism of simulated train movements and capacity analysis.

In this thesis, signalling is not fully modelled as an operational control system. However, it remains an important limitation and a key concept for interpreting the realism of the generated simulations. The absence or simplification of signalling means that the simulation can validate connectivity and route generation, but cannot fully reproduce all safety-critical operational constraints of a real railway network.

From a simulation perspective, signalling can be represented at different levels of detail. A simple simulation may ignore signalling and only check whether routes exist. A more realistic simulation must represent block occupation, signal states, route reservations, interlocking constraints, and train separation. In this thesis, signalling is not fully modelled as an operational control system. However, it remains an important limitation, since the generated simulations can validate connectivity and route generation but cannot fully reproduce all safety-critical operational constraints of a real railway network.

2.2.2 Railway Operations

Railway operations describe how trains are planned, dispatched, controlled, and managed over the infrastructure. While infrastructure defines what is physically possible, operations define what is actually allowed and scheduled. Railway operations combine timetables, train paths, rolling stock constraints, crew constraints, safety rules, capacity management, and real-time traffic regulation [38].

For a railway DT, operations are essential because the objective is not only to represent the network, but also to reproduce or analyse train movements. Operational data provides information about scheduled and actual train movements, delays, stops, and service patterns. This data can be used to generate simulation scenarios, validate the model, and study the behaviour of the railway system.

Train Movements

A train movement is the circulation of a train from one point of the network to another. It is defined by an origin, a destination, a route, a departure time, intermediate stops, and operational constraints. A train cannot freely choose its path at any time; its movement must be compatible with the infrastructure, timetable, signalling rules, and other train movements.

In simulation, train movements must be converted into routes and departure times. At the macro level, a route may be represented as a sequence of stations. At the micro level, it must be converted into a sequence of track segments or edges. This conversion is one of the central challenges of railway digital twin construction, because real operational records often describe train movements at station level, while the simulator requires paths through the generated network.

The feasibility of a train movement depends on both connectivity and timing. Connectivity means that a valid path must exist between consecutive locations. Timing means that the train must be inserted at the correct time and must respect operational constraints such as dwell times and travel times. If either the connectivity or timing is inconsistent, the simulation may fail or produce unrealistic behaviour.

Timetables

A railway timetable defines the planned movements of trains over the network. It specifies departure and arrival times, stopping patterns, running times, dwell times, and sometimes platform assignments or path allocations. Timetabling is a constrained optimization problem because it must satisfy service objectives while respecting infrastructure capacity, minimum headways, station capacity, rolling stock availability, and safety margins [26].

Periodic timetables are common in passenger railway systems. In such timetables, services repeat at regular intervals, for example every 30 or 60 minutes. This makes the service easier to understand for passengers and easier to coordinate across lines. However, periodicity also introduces constraints because connections, crossings, and resource usage must remain consistent across repeated time cycles.

A common way to visualize timetables is the time-space diagram. In such a diagram, one axis represents time and the other represents position along a railway line. Each train is represented by a line. The slope reflects speed, stops appear as horizontal segments, and crossings or overtakes can be analysed visually. Time-space diagrams are useful for understanding capacity, conflicts, delays, and recovery margins.

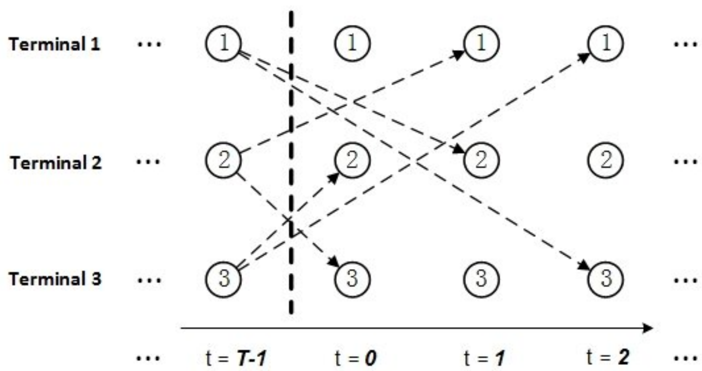

Figure 2.7: Example of a time-space diagram.

Capacity Management

Railway capacity refers to the ability of infrastructure to accommodate train movements within a given period while respecting safety, operational, and quality constraints. Unlike road capacity, railway capacity depends strongly on timetable structure, train heterogeneity, signalling, station dwell times, and route conflicts. It is therefore not only a property of the physical infrastructure, but also of how this infrastructure is used [26, 32, 23].

A railway line with homogeneous trains running at similar speeds and stopping patterns can usually accommodate more trains than a line where fast, slow, passenger, and freight trains are mixed. This is because heterogeneous traffic increases the need for headways, overtaking, and timetable margins. Similarly, a station with limited platforms or conflicting access routes may reduce capacity even if the surrounding tracks are not saturated.

The UIC 406 method is often used as a reference approach for railway capacity analysis. It evaluates capacity consumption by compressing train paths in a timetable and analysing how much time is occupied by train movements. This approach highlights that capacity is related not only to the number of trains, but also to their ordering, spacing, speed differences, and operational margins[27].

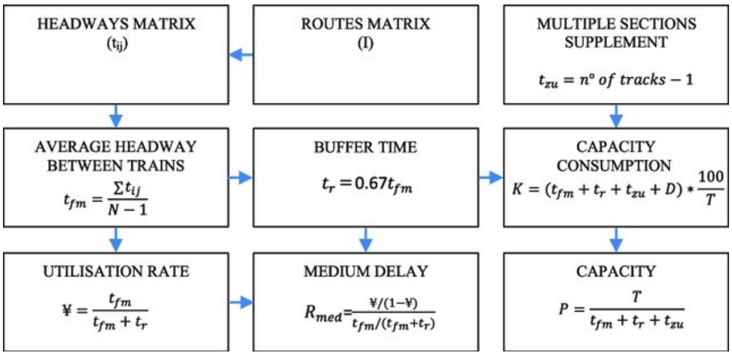

Figure 2.8: UIC 406 method procedure.

Capacity management is important for simulation because a generated route may be technically connected but operationally unrealistic if capacity constraints are ignored. For example, several trains may be assigned to the same platform or track at overlapping times. A detailed simulator should therefore account for infrastructure occupation, route conflicts, and safe separation. In this thesis, capacity is not the main optimization objective, but it remains relevant for interpreting the limitations of the macro-level and micro-level simulations.

Delay Propagation

Delay propagation is one of the most important phenomena in railway operations. A primary delay occurs when a train is delayed by an external or local cause, such as technical failure, passenger boarding time, weather, infrastructure problems, or operational incidents. A secondary delay occurs when this initial delay affects other trains through shared infrastructure, rolling stock circulation, crew dependencies, passenger connections, or timetable conflicts[21, 13].

Railway networks are particularly sensitive to delay propagation because many trains share the same tracks, stations, junctions, and timetable margins. A delayed train may occupy a track section later than planned, preventing another train from using it. It may also miss a planned crossing, delay a connection, or arrive late at a station where the rolling stock is needed for another service. As a result, a local disturbance can spread through the network.

Delay propagation can be studied at different levels. Microscopic approaches simulate train movements and conflicts in detail. Macroscopic approaches analyse delays at the level of stations, lines, or aggregated regions. Data-driven approaches use historical operational data to estimate how delays evolve over time and space. In this thesis, delay propagation is not fully modelled as a predictive process. However, reconstructing infrastructure and train movements is a prerequisite for such applications.

2.2.3 Railway Network Representation

A railway network can be represented in several ways depending on the modelling objective. The choice of representation determines which phenomena can be analysed and which assumptions are made. For digital twin and simulation purposes, three broad families of representations are particularly relevant: graph-based modelling, physics-based modelling, and logic-based modelling. These representations are not mutually exclusive, and detailed railway simulators may combine them.

Graph-Based Modeling

In graph-based modelling, the railway network is represented as a graph in which nodes correspond to stations, switches, junctions, or operational points, while edges correspond to track segments or connections between them [22]. This representation is compatible with algorithms from graph theory and operations research. It is particularly useful for routing, timetable design, resource allocation, connectivity analysis, and delay propagation studies.

The main limitation of graph-based models is that they often simplify train dynamics and operational constraints. If the graph is too abstract, trains may be treated as moving along edges without detailed acceleration, braking, signalling, or conflict constraints. Nevertheless, graph-based modelling is central to this thesis. The macro-level model uses a station-to-station graph, while the micro-level model builds a more detailed graph based on track segments and routing possibilities.

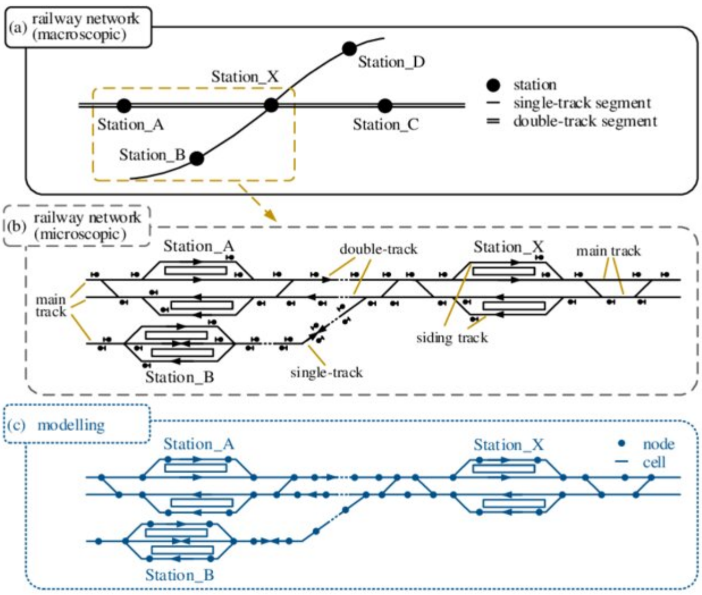

Figure 2.9: Example of a railway station modelling at macro and micro levels.

Physics-Based Modeling

Physics-based models represent the physical behaviour of trains and their interaction with the infrastructure. They simulate train motion as a function of traction force, resistance, braking, gradients, inertia, and sometimes energy consumption [18]. This approach is useful when the objective is to study train dynamics, energy-efficient driving, rolling stock performance, or precise running times.

The strength of physics-based modelling is its realism, but it requires detailed data such as train mass, traction characteristics, braking curves, gradients, and speed restrictions. It can also become computationally expensive at network scale.

Logic-Based Modeling

Logic-based models capture the discrete control rules and safety constraints of railway operations. A typical example is the modelling of interlocking systems, which ensure that only safe and conflict-free train movements are allowed. These models represent the railway as a set of states and transitions, with rules depending on the position of switches, signal states, block occupation, and route reservations.

Logic-based models are useful for validating signalling and safety-critical behaviour. They can be expressed using formal methods and model-checking tools such as UPPAAL or NuSMV [?, 2]. However, they require highly detailed rule definitions and may face scalability challenges when the network grows or when many trains operate simultaneously.

Each of the three modeling captures different but complementary aspects of a railway system. Combining graph abstraction for routing, physics modules for motion and simulation, and formal logic for safety are increasingly explored in the industry to connect these domains and support robust digital representations of railway networks.

Each modelling family captures a different aspect of the railway system. Graph-based models emphasize connectivity and routing, physics-based models emphasize motion and train dynamics, and logic-based models emphasize safety and operational rules. The work developed in this thesis mainly relies on graph-based representations, while acknowledging that more complete railway digital twins would require stronger integration of physics-based and logic-based constraints.

2.2.4 Railway Data Sources

Railway modelling and simulation depend strongly on the availability, completeness, and consistency of data. A digital twin requires data describing both the physical infrastructure and the operation of trains. These two categories are often stored in different systems, they can follow different formats, and use different levels of granularity. Aligning them is one of the main challenges of railway DT construction.

Infrastructure data describes what exists in the network, while operational data describes how the network is used. A simulation-oriented digital twin must combine both: infrastructure data is required to build the network, and operational data is required to generate train movements and validate the behaviour of the simulation.

Infrastructure Data

Infrastructure data includes information about tracks, switches, stations, platforms, signals, speed limits, electrification, geometry, and topology. Depending on the source, this data may be available as geospatial data, tabular data, XML files, engineering databases, or standardized railway formats.

OpenStreetMap and OpenRailwayMap are examples of open geospatial sources that can provide railway infrastructure information. They may include track geometry, station locations, railway line types, electrification, gauge, signals, and other attributes. These sources are useful because they are publicly accessible and cover large areas. However, their completeness and accuracy may vary depending on the region and the contributions of the mapping community.

OpenStreetMap and OpenRailwayMap are examples of open geospatial sources that can provide railway infrastructure information. They may include track geometry, station

locations, railway line types, electrification, gauge, signals, and other attributes. These sources are useful because they are publicly accessible and cover large areas [7]. However, their completeness and accuracy may vary depending on the region and the contributions of the mapping community.

Institutional data sources, such as infrastructure manager databases, are usually more accurate and complete but are often private or restricted. In Belgium, infrastructure data from Infrabel or related railway stakeholders can provide more reliable information about stations, tracks, platforms, and operational assets [1]. However, such data may require preprocessing, cleaning, transformation, and alignment before it can be used for simulation.

Standardized formats are also important for railway data exchange. RailML, GTFS and related railway modelling standards provide structured ways to describe infrastructure, rolling stock, timetables, and operational data [8, 6]. They improve interoperability between tools, but converting raw datasets into such formats can require significant modelling effort.

Operational Data

Operational data describes the planned or actual movement of trains. It may include timetables, train identifiers, station stops, arrival and departure times, delays, platform assignments, rolling stock information, and disruption records. Operational data is essential for transforming a static infrastructure model into a dynamic simulation scenario.

Timetable data provides the planned structure of train services. It defines which trains should run, when they should depart, where they should stop, and how they should connect to other services. Punctuality data provides information about actual train movements and deviations from the planned timetable. It can be used to analyse delay patterns, validate simulation results, or generate realistic train routes.

Operational data is often difficult to use directly. It may contain missing records, inconsistent station identifiers, cancelled trains, duplicated entries, or timing errors. It may also describe movements at station level, while the simulation requires detailed paths at track level. Therefore, cleaning and transformation are necessary before operational data can be used in a simulator.

A key challenge is the alignment between operational data and infrastructure data. A train record may indicate that a train travelled from one station to another, but the infrastructure model must contain a valid path between these locations. If the infrastructure is incomplete or if station identifiers do not match, the route cannot be generated. This challenge is particularly important in the micro-level model, where route generation depends on detailed track connectivity.

2.3 Railway Simulators

Simulation plays an important role in the analysis, planning, and evaluation of railway systems. Since railway operations involve many interacting components, it is often difficult to test new strategies directly on the real network. Simulators provide a controlled environment in which infrastructure layouts, timetables, train movements, disruptions, and operational scenarios can be evaluated before being applied in practice.

In the context of DTRs, simulation is particularly important because the virtual representation must be able to reproduce at least part of the behaviour of the physical system.

The required level of detail depends on the objective of the study. A high-level model may be sufficient for analysing station-to-station movements, while a detailed infrastructure model is required to study switches, platforms, signalling constraints, local conflicts, and route feasibility.

Railway simulators can be classified according to several criteria. A first distinction concerns the level of granularity. Macroscopic simulators represent traffic at an aggregated level, using stations, lines, flows, or timetable events. Microscopic simulators represent individual trains and infrastructure elements in more detail. A second distinction concerns the modelling objective: some tools focus on operational simulation, while others focus on infrastructure design, timetable planning, capacity analysis, or real-time traffic management. Finally, simulators differ in openness, extensibility, data requirements, interoperability, and support for automated workflows.

2.3.1 Overview of Existing Simulators

Several simulation platforms exist for railway and transport systems. Some are specifically designed for railways, while others are general-purpose or multimodal traffic simulation environments that can be adapted to rail use cases. Table [**?**] summarizes the main tools considered in this thesis.

Table 2.1: Comparative analysis of railway and transport simulation tools

Simulator	Type	Main focus	Main strengths	Main limitations	Relevance for this thesis
OpenTrack	Microscopic railway simulator	Railway operation, timetable evaluation, capacity and infrastructure analysis	Detailed railway-specific modelling; supports infrastructure, rolling stock, timetables and signalling constraints; provides train graphs, occupation diagrams and operational statistics	Requires detailed and well-structured railway data; less suited for fully custom open data pipelines; not primarily designed as an open experimental framework	Highly relevant as a reference railway simulator, but less suitable for the implemented data-driven and reproducible pipeline

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Table 2.1 – continued from previous page

Simulator	Type	Main focus	Main strengths	Main limitations	Relevance for this thesis
RailSys	Professional railway planning and simulation suite	Capacity analysis, timetable planning, travel time computation, headway analysis and ETCS-related studies	Strong railway-specific capabilities; designed for infrastructure managers and railway operators; supports detailed operational and capacity studies	Commercial/professional environment; limited openness for academic reproducibility; requires detailed infrastructure and operational input data	Relevant for comparison with professional railway tools, but less aligned with the objective of building an open and programmable prototype
OSRD	Open-source railway-specific platform	Railway infrastructure design, capacity analysis, timetabling, simulation and short-term path requests	Open-source; railway-oriented by design; supports infrastructure and timetable workflows; suitable for capacity and pathfinding studies	More oriented toward railway design and planning workflows; integration with a custom heterogeneous data-processing pipeline may require additional modelling effort	Relevant because it is open-source and railway-specific, but SUMO was preferred for its flexibility, scriptability and simulation workflow control
SUMO	Open-source microscopic traffic simulator	Multimodal traffic simulation, route generation, demand modelling, public transport and railway simulation	Open-source and highly scriptable; supports XML-based network generation, routing tools, outputs, TraCI interaction and large-scale automated workflows; includes railway-specific features such as rail edges, train stops and rail signals	Not originally designed as a railway-only simulator; detailed signalling, interlocking, dispatching and platform assignment require additional modelling effort	Selected as the main simulation platform because it enables full control over network generation, trip generation, execution and output processing

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Table 2.1 – continued from previous page

Simulator	Type	Main focus	Main strengths	Main limitations	Relevance for this thesis
AnyLogic / MATSim	General-purpose or transport simulation platforms	Agent-based simulation, multimodal mobility, large-scale transport demand modelling	Useful for multimodal and agent-based transport scenarios; suitable for studying passenger behaviour, demand patterns or large-scale mobility systems	Limited native support for detailed railway infrastructure constraints such as switches, interlocking, block occupation and platform-level operations	Useful as general transport simulation references, but less appropriate for the railway infrastructure reconstruction objectives of this thesis

The choice of a simulator depends on the intended use case. For a railway digital twin foundation, several criteria are important. The simulator must represent the infrastructure at the required level of detail, support train route generation and execution, be compatible with available data sources, produce outputs that can be compared with operational data, and allow reproducible experimentation.

Specialized railway simulators such as OpenTrack and RailSys provide strong railway-specific capabilities. They are designed to represent timetables, train movements, signalling systems, capacity constraints, and detailed infrastructure. Their main advantage is that they directly encode railway concepts that are simplified or absent in general-purpose traffic simulators. However, they often require detailed infrastructure and timetable data, and their integration into a fully custom data-processing pipeline can be less straightforward.

OSRD is an interesting alternative because it is both open-source and railway-specific. It directly targets railway infrastructure design, capacity analysis, timetabling, and pathfinding. However, its data model is more oriented toward railway planning workflows, whereas this thesis focuses on transforming heterogeneous infrastructure and operational data into simulation-ready artefacts within a flexible and programmable environment.

SUMO occupies a different position. It is less railway-specialized than OpenTrack, RailSys, or OSRD, but it provides a highly flexible simulation environment. Its main strengths are openness, scalability, scriptability, and the availability of tools for network generation, routing, demand generation, output processing, and online interaction. These characteristics make SUMO suitable for experimental research where the objective is not only to run a simulation, but also to generate the simulation network automatically from data, modify scenarios programmatically, and analyse outputs with custom scripts.

For this thesis, SUMO was selected because it allows the complete simulation workflow to be controlled. Networks can be generated from XML files, routes can be created automatically, simulations can be executed through command-line tools, and outputs can be parsed into datasets suitable for analysis. This level of control is essential for comparing macro-level and micro-level railway representations and for evaluating the feasibility of integrating real operational data into a simulation pipeline.

2.3.2 Limitations and Gaps in Current Simulators

Existing simulators provide important capabilities, but several limitations remain when the objective is to construct a railway digital twin from heterogeneous real-world data. A first limitation concerns data integration. Many simulators assume that infrastructure, timetable, and demand inputs are already available in a consistent and simulator-compatible format. In practice, railway data often comes from multiple sources, with different identifiers, geometries, levels of detail, and temporal resolutions. Transforming these datasets into valid simulation inputs remains a major challenge.

A second limitation concerns the balance between realism and scalability. Detailed railway simulators can represent signalling, switches, platforms, and operational conflicts, but this usually requires high-quality infrastructure data and careful configuration. Simpler models are easier to generate and scale, but they omit important operational constraints. This creates a trade-off between macroscopic representations, which are easier to use with large datasets, and microscopic representations, which are more realistic but more sensitive to data quality and topological inconsistencies.

A third limitation concerns reproducibility and openness. Some railway simulation tools are commercial or rely on proprietary workflows. This can be appropriate for industrial applications, but it may limit academic reproducibility. Open-source tools such as SUMO and OSRD reduce this barrier, but they still require significant preprocessing and modelling work before real railway data can be used.

A final limitation concerns the connection between simulation and operational data. Many simulation studies focus either on infrastructure modelling or on timetable evaluation, but fewer provide an end-to-end workflow that starts from raw infrastructure and punctuality data, generates a simulation-ready railway network, executes train movements, and compares simulation outputs with real observations. This gap directly motivates the work presented in this thesis.

2.3.3 SUMO: General Features

SUMO is an open-source traffic simulation suite designed to model large-scale transport systems. It is microscopic, meaning that each vehicle is represented individually, and time-continuous, meaning that vehicle movements evolve during the simulation rather than being represented only as isolated events. SUMO supports several transport modes, including cars, buses, bicycles, pedestrians, public transport vehicles, and rail vehicles [31, 16].

A SUMO scenario is built from several input files. The network file describes the topology of the simulated environment, including nodes, edges, lanes, connections, and permissions. Route files define vehicles, vehicle types, departure times, and paths. Additional files may define stops, detectors, traffic lights, calibrators, rerouting rules, public transport stops, or other simulation objects. A configuration file then specifies which inputs must be loaded and which outputs must be generated.

SUMO also provides several tools for scenario creation and manipulation. `netconvert` converts network descriptions into SUMO network files and can import networks from different formats, including OpenStreetMap and manually defined XML files. `duarouter` computes routes from trips or origin-destination information. `randomTrips.py` generates synthetic travel demand. `netedit` provides a graphical interface for editing networks. These tools make SUMO suitable for automated simulation pipelines.

SUMO also provides a rich set of outputs. Depending on the configuration, it can generate information about completed trips, stops, vehicle trajectories, edge-based statistics, lane-based statistics, emissions, delays, and simulation summaries. These outputs are useful because they can be transformed into structured datasets and compared with real observations. In this thesis, outputs such as trip information, stop information, and summary statistics are particularly important because they allow simulated train movements to be evaluated against operational data.

Another important feature is the availability of structured outputs. SUMO can generate trip information, stop information, vehicle trajectories, edge-based statistics, lane-based statistics, emissions, summary files, and other outputs depending on the configuration. These files can be parsed after simulation and converted into analysis-ready datasets. For this thesis, this is particularly important because the simulation outputs must be compared with railway operational data.

2.3.4 SUMO Backend Architecture

SUMO follows a modular architecture built around XML input files, command-line tools, simulation executables, and optional programming interfaces. This architecture makes it possible to generate, execute, and analyse simulations automatically.

The main simulation executable, `sumo`, runs scenarios without a graphical interface and is therefore suitable for batch experiments. The graphical version, `sumo-gui`, provides visual inspection of the simulation, which is useful for debugging network geometry, train routes, stops, and conflicts. The simulation kernel loads the network, vehicles, routes, stops, and configuration files, then updates the state of each vehicle during the simulation.

Network generation is handled by dedicated tools such as `netconvert` and `netgenerate`. In the context of this thesis, `netconvert` is especially important because it transforms node and edge definitions into a complete SUMO network. This step includes the construction of lanes, connections, permissions, and network topology. For railway simulation, this conversion step is critical because routing feasibility depends on the correctness of the generated topology.

Demand generation and routing are handled by tools such as `randomTrips.py` and `duarouter`. `randomTrips.py` can generate synthetic trips according to user-defined parameters, while duarouter computes valid paths over the network. In railway scenarios, these tools can be used to generate artificial train movements or to convert reconstructed operational trips into executable routes.

SUMO also includes Python libraries such as `sumolib`, which allows SUMO networks and XML files to be inspected and manipulated programmatically. This is important for research workflows because it makes it possible to build custom preprocessing, validation, and post-processing scripts. Together with `TraCI`, these libraries provide a bridge between SUMO and external data-processing environments.

Several auxiliary tools support the simulation workflow. `netconvert` is used for network generation, `duarouter` for route computation, `randomTrips.py` for demand generation, and `netedit` for manual inspection or editing. In addition, `sumolib` provides Python utilities for reading and processing SUMO networks and output files. `TraCI`, the Traffic Control Interface, allows external programs to interact with a running simulation by retrieving vehicle states, modifying routes, changing parameters, or controlling elements during execution.

This modular design is useful for DT development because each stage of the pipeline can be automated. Infrastructure data can be transformed into nodes and edges, operational data can be transformed into train trips, routing tools can compute paths, and output files can be converted into datasets for validation.

2.3.5 SUMO For Railway Simulation

Although SUMO was originally developed as a general traffic simulator, it includes features that can be adapted to railway simulation. Rail vehicles can be represented using vehicle types and route definitions. Railway infrastructure can be encoded through edges and lanes with rail permissions. Train stops can be represented using stopping areas attached to lanes, and schedules can be defined through departure times and planned stops. SUMO also provides support for public transport lines, rail signals, crossings, and bidirectional track usage in certain modelling configurations [16, 15].

For railway use cases, SUMO offers two important advantages. First, it can represent the network at different levels of abstraction. At the macro level, stations can be represented as nodes and station-to-station links as rail edges. At the micro level, individual track segments, platforms, and switches can be represented more explicitly. Second, SUMO supports automated scenario generation. This makes it possible to generate railway simulations from data rather than manually constructing each scenario.

However, SUMO also has limitations for railway simulation. Detailed signalling, interlocking, dispatching, platform assignment, block occupation, and conflict resolution are not automatically inferred from raw infrastructure data. They must either be simplified, explicitly encoded, or controlled through additional logic. Therefore, SUMO is well suited for building a flexible simulation-oriented digital twin foundation, but it does not remove the need for careful railway modelling.

In this thesis, SUMO is used as a simulation backend rather than as a complete railway planning system. The main objective is to evaluate whether heterogeneous infrastructure and operational datasets can be transformed into executable railway simulations. SUMO is therefore appropriate because it offers full control over the generation of networks, routes, stops, configuration files, and outputs. Its limitations are also useful from a methodological perspective, because they reveal which railway-specific elements must be explicitly reconstructed before a more complete digital twin can be achieved.

2.4 Simulation Calibration

Simulation calibration is the process of adjusting model parameters so that simulation outputs reproduce observed real-world behaviour as closely as possible. In transport simulation, calibration is necessary because a simulator is never a perfect copy of reality. It relies on assumptions, simplifications, parameters, and incomplete data. Without calibration, a simulation may be technically executable while still producing unrealistic travel times, traffic flows, delays, or capacity usage.

In microscopic traffic simulation, calibration often concerns parameters related to vehicle behaviour, such as acceleration, deceleration, reaction time, lane-changing behaviour, route choice, and departure distributions. In railway simulation, the relevant parameters are different. Calibration may concern running times, dwell times, acceleration and braking behaviour, platform stopping times, route choices, dispatching rules, delay distributions, and capacity constraints.

Several calibration approaches can be used. Manual calibration relies on expert knowledge and compares simulation outputs with reference data. It is simple and interpretable, but can be subjective and difficult to scale. Optimization-based calibration defines an objective function measuring the difference between simulated and observed data, then searches for parameter values that minimize this difference. Data-driven calibration uses statistical or machine learning methods to estimate parameters directly from historical observations.

A calibration process generally requires three elements: reference data, comparable simulation outputs, and evaluation metrics. In railway applications, reference data may include planned and observed arrival times, departure times, dwell times, delays, platform usage, or track occupation. The corresponding simulation outputs can then be compared using metrics such as mean absolute error, root mean square error, travel time error, delay distribution similarity, punctuality indicators, or differences between observed and simulated traffic volumes.

SUMO includes calibration-related mechanisms, especially for road traffic. For example, calibrators can insert or remove vehicles in order to match observed flow and speed values over specific time intervals [14]. More generally, SUMO can also be integrated into external calibration workflows because simulations can be launched automatically, parameters can be modified through scripts, and outputs can be analysed programmatically.

In this thesis, calibration is not the main objective. The purpose is not to tune the simulator until it reproduces Belgian railway operations with high precision, but to construct simulation-ready railway models and evaluate whether heterogeneous infrastructure and operational data can be transformed into feasible SUMO scenarios. Nevertheless, calibration remains important for future work. Once infrastructure reconstruction and route feasibility issues are addressed, calibration could be used to adjust dwell times, running times, departure distributions, and delay behaviour so that the simulated railway system better matches historical punctuality data.

2.5 Research Gaps

The literature shows that digital twins and railway simulators provide a strong foundation for railway analysis, but several gaps remain when the objective is to construct a simulation-oriented railway digital twin from heterogeneous real-world data.

The first gap concerns data transformation. Existing simulation tools often assume that infrastructure, timetable, and operational data are already available in consistent and simulator-compatible formats. In practice, railway datasets differ in structure, identifiers, spatial precision, temporal granularity, and level of detail. Infrastructure data may describe stations, tracks, switches, platforms, or geometries, while operational data may only describe train movements at station level. Transforming these heterogeneous datasets into valid simulation inputs is therefore a central challenge.

The second gap concerns the level of network representation. Macro-level models, where stations are represented as nodes and railway connections as edges, are easier to generate and more robust for large-scale simulation. However, they ignore detailed infrastructure constraints such as platforms, switches, parallel tracks, and station-level routing. Micro-level models are closer to the physical infrastructure, but they are more sensitive to missing connections, incorrect topology, platform assignment errors, and routing inconsistencies. The practical trade-off between macro-level robustness and micro-level realism remains insufficiently explored.

The third gap concerns the integration of real operational data. Synthetic trips can test network connectivity, but they do not reproduce real railway operations. Punctuality data provides a more realistic basis, but it does not contain the exact physical route followed by each train. Train journeys must therefore be reconstructed from station sequences and mapped onto the generated network. This is manageable at the macro level, but more difficult at the micro level, where each movement must correspond to a valid sequence of track segments and platform stops.

This micro-level difficulty is especially important in this thesis. The failure to reconstruct some real train routes is not only an implementation limitation, but also a methodological finding. It shows that high infrastructure coverage does not necessarily imply route feasibility. A reconstructed network must also be topologically consistent, and the dual graph used for routing must preserve valid transitions between track segments. Missing connections, wrong directions, inaccurate platform matching, or excessive segmentation can prevent real station sequences from becoming executable routes.

This thesis addresses these gaps by developing a SUMO-based pipeline at two levels of abstraction. The macro-level model evaluates the integration of real operational data into a simplified station-to-station simulation. The micro-level model evaluates the feasibility and limitations of detailed infrastructure reconstruction based on tracks, switches, platforms, and routing structures. Together, these models make it possible to analyse the trade-off between simplicity, scalability, realism, and simulation feasibility.

Chapter 3

Proposed Methodology

This chapter presents the methodology used to construct a simulation-oriented digital twin of a railway network. The objective is not to build a complete real-time digital twin, but to define the main steps required to transform heterogeneous railway datasets into SUMO-compatible simulation models.

The proposed approach is based on two complementary levels of representation. The macro-level model represents the railway network as a station-to-station graph and is mainly used to integrate real operational data into simulation. The micro-level model represents the infrastructure in more detail through tracks, switches, platforms, and routing structures. This distinction makes it possible to compare a simplified and robust representation with a more detailed but more demanding one.

3.1 Positioning of This Work

The state of the art shows that railway digital twins and simulation tools provide useful foundations for analysing railway systems. However, constructing an executable simulation from real railway data remains difficult. Infrastructure data and operational data are often heterogeneous, incomplete, and represented at different levels of detail. Existing simulators also generally assume that the required input files are already available in a consistent and simulator-compatible format.

This thesis addresses this gap by focusing on the transformation of railway data into simulation-ready artefacts. The contribution is therefore not the development of a new simulator, but the design and evaluation of a data-driven pipeline that connects infrastructure reconstruction, operational data integration, route generation, simulation execution, and output analysis.

The work is positioned around the comparison of two modelling levels. The macro-level model evaluates how far real punctuality data can be integrated into a simplified station-to-station railway simulation. The micro-level model investigates whether a more detailed infrastructure representation can support realistic railway routing based on tracks, platforms, switches, and a reconstructed dual graph. This comparison is central to the thesis because it highlights the trade-off between abstraction, scalability, infrastructure realism, and simulation feasibility.

3.2 Framework

The proposed framework follows a data-driven pipeline that converts raw railway datasets into SUMO-compatible simulation scenarios. The pipeline is composed of four main stages: data preprocessing, infrastructure modelling, trip generation, and simulation evaluation.

First, the available railway datasets are cleaned and standardized. Infrastructure and operational data must be aligned because they do not always share the same identifiers, formats, or level of detail. This step includes filtering the selected simulation area, harmonizing station information, processing geographical coordinates, and converting temporal values into simulation-compatible timestamps.

Second, the railway infrastructure is reconstructed according to the selected level of abstraction. At the macro level, stations are represented as nodes and railway links as edges. This graph is suitable for reconstructing station-to-station train movements from punctuality records. At the micro level, the infrastructure is represented through individual track segments, detected switches, platform-track associations, and routing structures. This representation is closer to the physical infrastructure, but it requires stronger topological consistency.

Third, train movements are generated. In the macro-level model, trips are reconstructed from real punctuality records by grouping train passages by train identifier and operating day, then sorting them chronologically. In the micro-level model, two strategies are considered: synthetic trip generation and real-data-based trip reconstruction. Synthetic trips are used to test network connectivity, while real-data trips aim to reproduce observed train movements on the detailed infrastructure.

Finally, the generated networks and routes are executed in SUMO. The simulation outputs are then processed into structured datasets that can be analysed and compared with the expected behaviour of the model. This evaluation focuses on the quality of the generated infrastructure, the feasibility of train routing, and the ability of each representation to support railway simulation.

3.3 Why SUMO?

SUMO was selected because the objective of this thesis requires a flexible, open-source, and scriptable simulation platform. Specialized railway simulators such as OpenTrack, RailSys, or OSRD provide strong railway-specific functionalities, but they often require detailed and well-structured railway input data. In contrast, this work focuses on building a custom pipeline from heterogeneous datasets, which requires full control over network generation, route generation, execution, and output processing.

SUMO was selected because it is open-source, scriptable, and compatible with automated data-processing workflows. Its XML-based input structure makes it possible to generate networks, routes, stops, and configuration files directly from Python scripts. It also provides command-line tools which are useful for converting networks, generating trips, and computing executable routes.

SUMO is well suited to this objective because its XML-based structure allows networks, routes, stops, vehicle types, and configuration files to be generated automatically from Python scripts. It also provides command-line tools such as `netconvert`, `randomTrips.py`, and `duarouter`, which support automated network conversion, synthetic trip generation, and route computation.

Another advantage is that SUMO can support both levels of representation considered in this thesis. At the macro level, stations can be represented as nodes and station-to-station connections as railway edges. At the micro level, more detailed infrastructure can be represented through rail edges, lanes, train stops, and route definitions. This flexibility makes SUMO suitable for comparing simplified and detailed railway simulation approaches within the same environment.

However, SUMO is not a railway-only simulator. Detailed signalling, interlocking, dispatching, platform assignment, and conflict management are not automatically inferred from raw data. These limitations must be taken into account when interpreting the simulations. For this thesis, this trade-off is acceptable because the main objective is to study how railway data can be transformed into simulation-ready models, rather than to reproduce all operational rules of a professional railway simulator.

3.4 Data Available

The methodology relies on two main categories of data: infrastructure data and operational data. Infrastructure data describes the railway network itself, while operational data describes how trains move through this network. Both are required to construct a simulation-oriented railway digital twin.

The available data is not directly usable by SUMO. It must first be cleaned, filtered, transformed, and converted into simulation files. The type of processing required depends on the level of representation. The macro-level model mainly uses stations and station-to-station links, whereas the micro-level model requires track geometry, platforms, switches, and routing structures. The data used in this thesis comes from Infrabel Open Data Portal And DTSC who has access to private data directly from Infrabel.

3.4.1 Infrastructure Data

Infrastructure data describes the static railway network. It is used to generate the physical or topological environment in which trains circulate. The macro-level and micro-level models use different infrastructure datasets because they represent the railway system at different granularities.

At the macro level, the network is represented as a graph of stations and direct connections. This abstraction is sufficient for simulating train movements between neighbouring stations, but it does not include platforms, switches, signals, or detailed station layouts.

Dataset	Example attributes	Reference value	Role in Methodology
Stations	Station identifier, station name, coordinates	559 stations	Create SUMO nodes
Stations Connections	Origin, destination, distance, geometry	1385 tracks	Create SUMO railway edges
Selected Simulated Area	List of neighbouring stations	User-defined	Restrict the simulation scope

Table 3.1: Macro-Level Infrastructure Data Overview

At the micro level, the infrastructure is represented in more detail. Track segments are used to build the railway graph, switches are inferred from the topology, and platforms are associated with nearby track segments. This representation is necessary for more realistic routing, but it is more sensitive to missing connections or incorrect topology.

Dataset	Example attributes	Reference value	Role in Methodology
Tracks segments	Track identifier, geometry, length	25068 tracks	Build detailed railway edges
Stations	Station identifier, station name, coordinates	554 stations	Support platform detection
Platforms	Station name, platform number, station's position	1560 platforms	Create SUMO train stops
Switches	Node identifier, coordinates, connected tracks	10379 switches	Support routing and topology analysis

Table 3.2: Micro-Level Infrastructure Data Overview

3.4.2 Operational Data

Operational data describes train movements over the railway network. In this thesis, this information is contained in a single punctuality dataset. Each record corresponds to the passage of a train at a station and includes information related to train identification, station passage, planned times, observed times, and delays.

This dataset is used to reconstruct ordered train journeys. For each train and operating day, station records are grouped and sorted chronologically. At the macro level, these sequences can be mapped onto the station-to-station graph. At the micro level, they must be mapped onto a detailed path composed of tracks, platforms, and valid connections, which makes the process more complex.

Dataset	**Main attribute groups**	**Example attributes**	**Reference value**	**Role**
Punctuality dataset	Train identification, station passage, timetable information, observed operation, delays	Train identifier, operating date, station identifier, planned arrival/departure time, observed arrival/departure time, arrival delay, departure delay	Depends on selected time window	Reconstruct real train journeys and generate simulation routes

Table 3.3: Operational Data Overview

Overall, the available data provides the basis for constructing both macro-level and micro-level railway simulations. The infrastructure data defines the network to be simulated, while the operational data defines the train movements to reproduce. The central methodological challenge is therefore to align these datasets at the appropriate level of detail and convert them into valid SUMO inputs.

Chapter 4

Implementation

4.1 Macro-Level Modeling

The macro-level model constitutes the first operational implementation of the simulation-oriented railway digital twin developed in this thesis. Its objective is to generate a simplified but executable representation of the Belgian railway network in SUMO. At this level of abstraction, stations are represented as nodes and railway connections between stations as edges. Detailed infrastructure elements such as platforms, switches, individual tracks, signals, and station-level routing constraints are intentionally omitted.

This abstraction makes the model computationally manageable and suitable for validating the complete workflow, from raw railway data to simulation outputs. It also provides a baseline against which the more detailed micro-level model can later be compared. The implementation was developed in Python, using Apache Spark for data processing and SUMO for simulation execution. The workflow consists of four main stages: data processing, network generation, trip generation, and simulation.

4.1.1 Data Processing Pipeline

The macro-level model starts with a data processing pipeline that transforms infrastructure and operational datasets into a format compatible with SUMO. Three main types of data are used: station data, station-to-station connectivity data, and punctuality data. Station data provides identifiers, names, and geographical coordinates. Connectivity data describes the railway links between neighbouring stations. Punctuality data contains operational records of train passages through stations.

The preprocessing follows an Extract-Transform-Load approach. First, the raw datasets are loaded into Apache Spark dataframes. Spark is used mainly because the punctuality dataset can be large and requires efficient filtering, grouping, and temporal transformations. The transformation step then standardizes identifiers, processes geographical information, removes inconsistent records when necessary, and prepares the data for simulation generation.

A filtering step is also applied to restrict the simulation to the selected area and time period. The user can choose a subset of stations to simulate, and neighbouring stations required to preserve connectivity may be included automatically. This makes it possible to focus on coherent station chains rather than the entire national network, which reduces complexity while preserving meaningful train movements.

The output of this pipeline is a set of cleaned and harmonized datasets containing station information, station-to-station links, and train circulation records. These datasets are then used to generate the SUMO network and the corresponding train routes.

4.1.2 Railway Network Generation

After preprocessing, the railway network is generated as a station-to-station graph. Each station becomes a node, and each direct railway connection becomes an edge. The graph can be defined as $G = (V, E)$, where V is the set of stations and E is the set of railway links between them.

For each station, the implementation stores the station identifier, name, and geographical coordinates. For each connection, it stores the origin station, destination station, distance, and, when available, the geometry of the railway corridor. Preserving the geometry allows the generated SUMO network to remain visually close to the real railway layout, while keeping the model much simpler than the actual infrastructure.

The graph is then converted into SUMO-compatible XML files. The node file defines the stations included in the simulation, while the edge file defines the rail links between them. Additional files define train stops, vehicle types, route information, and the simulation configuration. The final network is generated with netconvert, which transforms the node and edge definitions into a SUMO network file.

This representation is simple, robust, and easy to generate. However, because each station is represented as a single node, the model cannot reproduce internal station operations such as platform allocation, switch occupation, signal interactions, or local route conflicts.

4.1.3 Trip Generation

Once the macro-level network has been generated, train movements are reconstructed from the punctuality dataset. The objective is to use real operational data rather than relying only on synthetic traffic.

The punctuality dataset contains station-level records for train passages. Each record includes a train identifier, a station identifier, and temporal information such as planned and observed times. However, the dataset does not directly provide complete SUMO routes. A reconstruction step is therefore required.

For each train and operating day, the records are grouped and sorted chronologically according to the planned timetable. Consecutive station pairs are then extracted to reconstruct the sequence of stations visited by the train. These station sequences are mapped onto the macro-level network and converted into SUMO route definitions.

Departure times must also be converted into SUMO simulation time. Since SUMO represents time as seconds elapsed since the start of the simulation, each planned departure time is transformed by computing its difference from the selected simulation start date.

This approach preserves realistic train movements, station usage patterns, and departure time distributions. It therefore provides a more representative simulation scenario than purely random traffic generation, while remaining compatible with the simplified station-to-station model.

4.1.4 Simulation

After the network and routes have been generated, the macro-level scenario can be executed in SUMO. The simulation configuration references all required files: the network file, route file, additional files for stops and trains, and output definitions.

The macro-level simulation is not intended to reproduce the full Belgian railway network in a single scenario. Although the generated network can theoretically include a large number of stations, simulating the complete national traffic would introduce a high level of complexity in capacity management in frequently visited stations (stations with dense traffic, multiple platforms, many intersecting services, and complex internal routing would require infrastructure details that are not represented at this level of abstraction). For this reason, the simulation is restricted to selected parts of the network (chosen by the user). These parts correspond to chains of neighbouring stations where the train traffic remains moderate. In practice, this means that the user selects a set of stations to be included in the simulation, and the generation pipeline constructs the corresponding local railway subnetwork.

By focusing instead on station chains with lower traffic intensity, the simulation remains coherent with the assumptions of the macro-level model. The simplified representation is therefore used in a context where local infrastructure conflicts are less dominant and where station-to-station train movements can still be meaningfully simulated.

The duration of the simulation is configurable. The user can specify the number of operating days to be included in the scenario. This parameter determines the time window used when filtering the operational data and therefore controls the number of train movements inserted into the simulation. A short simulation period can be used for debugging and validation, while a longer period can be selected to analyse traffic behaviour over multiple days.

Once the network, train routes, stop definitions, and simulation configuration files have been generated, the simulation can be launched directly in SUMO. During execution, trains circulate along the station-to-station graph according to the reconstructed routes and departure times. Since detailed infrastructure is not represented, train movements are mainly governed by the generated routes, travel times, and SUMO's internal simulation rules.

Several outputs are produced by the simulation. The `tripinfo` file provides information about completed train journeys, the `stopinfo` file records stopping events at stations, and `summary` outputs describe the global evolution of the simulation. A post-processing module parses these XML outputs and converts them into CSV files whose structure is closer to the original punctuality data.

This post-processing step reconstructs simulated arrival and departure information for each train and station. As a result, the macro-level simulation can be compared with historical operational data and used to evaluate whether the generated digital twin reproduces station-to-station train movements in a coherent way.

The macro-level workflow can therefore be summarized as follows: raw data is cleaned and standardized, a station-to-station SUMO network is generated, real punctuality records are transformed into train routes, the scenario is simulated, and the outputs are converted into datasets suitable for validation.

4.1.5 Limitations

Although the macro-level model validates the overall pipeline, it remains limited by its level of abstraction. Stations are represented as single nodes and railway corridors as direct edges. As a result, detailed infrastructure elements such as platforms, switches, signals, and individual tracks are not explicitly represented. The model therefore cannot accurately reproduce local operational constraints, platform conflicts, switch occupation, or station-level routing decisions.

The model is also dependent on the quality of the available infrastructure and operational data. It can reconstruct train movements at station level, but it cannot identify the exact physical route followed by trains inside complex railway areas. Nevertheless, the macro-level implementation remains an important first step: it demonstrates that real operational data can be integrated into a SUMO-based railway simulation and provides a robust baseline for comparison with the micro-level model.

4.2 Micro-Level Modeling

The micro-level model was developed to overcome the main abstraction of the macro-level representation. Instead of modelling the railway network only as stations connected by aggregated links, the micro-level model aims to represent the physical infrastructure in greater detail. Tracks, platforms, switches, and operational segments are explicitly reconstructed and converted into SUMO-compatible elements.

The objective of this model is to evaluate whether a more realistic railway simulation can be generated from the available infrastructure and operational datasets. Compared with the macro-level model, this approach provides a richer representation of the railway network, but it also introduces stronger requirements on data quality and topological consistency. A valid simulation requires not only that infrastructure elements are detected, but also that they form a connected and routable network.

This approach significantly increases the realism of the generated digital twin. The explicit representation of railway assets enables more accurate train movements, better infrastructure utilization, and the possibility of evaluating local operational phenomena such as conflicts at junctions, platform occupancy, and the propagation of delays within complex stations.

The implementation follows five main stages: data processing, infrastructure modelling, trip generation, simulation, and limitation analysis.

4.2.1 Data Processing Pipeline

The micro-level data processing pipeline follows an ETL process. The raw infrastructure data contains geographical track geometries represented as sequences of coordinates. These geometries are first cleaned and normalized in order to obtain a consistent spatial representation of the railway network.

Several processing steps are then applied to enrich the infrastructure. First, switches are detected from the topology of the track network. Then, platforms are associated with nearby track segments using a spatial proximity-based method. When several stations or platforms are assigned to the same track segment, a segmentation step is applied to increase the granularity of the network. Finally, the processed infrastructure is converted into graph structures used for route generation and simulation.

The output of the pipeline consists of a detailed railway network containing track segments, detected switch-like nodes, platform-to-track associations, and routing structures. These elements are then exported into SUMO files and used to generate train movements.

4.2.2 Infrastructure Modeling

Switch Detection

Switch detection is based on the topology of the railway graph. Track extremities are represented as nodes, and track sections are represented as edges. For each node, the number of connected track segments is computed.

Nodes connected to two track segments are considered part of a continuous railway line. Nodes connected to three or more track segments are classified as switch-like elements because they represent branching or merging points in the infrastructure. These detected switches are then used to enrich the micro-level network and support routing through the reconstructed topology.

The resulting switch and edge datasets forms the foundation of the micro-level network model and is subsequently used for graph construction, route generation, and simulation.

This approach does not require explicit switch labels in the source data. However, it depends strongly on the quality of the reconstructed track topology. Missing, duplicated, or incorrectly connected track segments can affect the number of detected switches.

Platform Detection

Platform detection aims to associate station platforms with the track segments on which trains can stop. The available data provides station locations and platform information, but these elements are not directly linked to SUMO lanes. A matching procedure is therefore required.

The implemented method relies on spatial proximity. For each station, a search area is defined around its coordinates. Nearby track segments are identified and ranked according to their distance from the station. If a station has N platforms, the N closest candidate track segments are selected and associated with the station platforms.

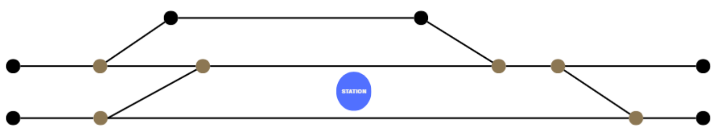

Figure 4.1: Example of platform detection initial setup

For each station, a research zone is defined around its coordinates. This research zone is a circular area with an adaptative radius (e.g., 100 meters) that encompasses the area where they are likely located. The radius of the research zone can be adjusted based on the density of platforms in the area, ensuring that it is large enough to capture all relevant track segments while minimizing the inclusion of irrelevant segments.

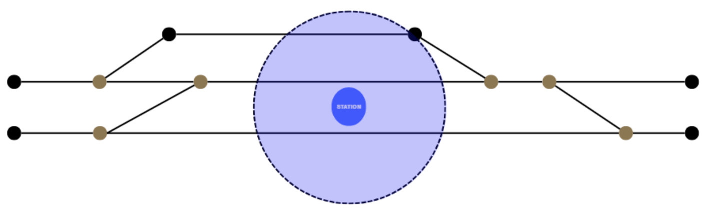

Figure 4.2: Example of platform detection searching zone

Within the research zone, the script identifies all track segments from the track dataset that intersect with the zone. Then, the script ranks the track segments based on their proximity to the station coordinates. The remaining track segments are classified into three categories (given a station that has n platforms):

1. Best candidates: the track segments that are closest to the station coordinates and are likely to correspond to a platform ($rank \leq n$).

2. Mediocre candidates: the track segments that are within the research zone (or close to it) but are not the best candidate. These segments may correspond to additional platforms or other track segments near the station ($n \leq rank \leq 1.5n$).

3. Irrelevant segments: the track segments that are outside the research zone or are too far from the station coordinates. These segments are unlikely to correspond to platforms and can be disregarded in the platform detection process ($1.5n \leq rank \leq 2n$).

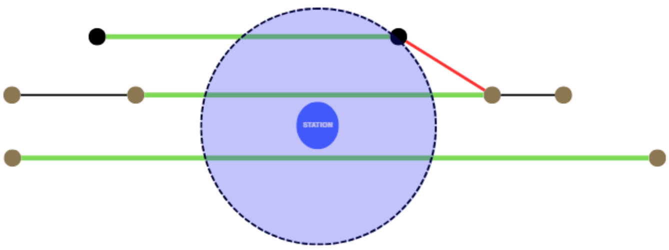

Figure 4.3: Example of platform detection ranking when the number of platform is 3

This method is simple and scalable, but it remains an approximation. It works well when platforms are located close to the corresponding tracks, but it can be less reliable in complex stations where many parallel tracks are located close to each other.

Track Segmentation

Track segmentation is introduced to improve the granularity of the reconstructed infrastructure. In some cases, a single track segment may be associated with several stations or platforms. This creates ambiguity because the same segment cannot represent several distinct stopping locations accurately.

To address this issue, long or conflicting track segments are split into smaller sub-segments while preserving the original geometry. After segmentation, the platform detection process can be applied again on a more detailed infrastructure representation. This improves the association between platforms and tracks and produces a network better suited for simulation.

First, all segments assigned to more than one station are identified. Only these conflicting segments are processed, while the remaining network remains unchanged. Each selected segment is then divided into smaller sub-segments according to a maximum segment length threshold of 800 meters.

New sub-segments are created exclusively using the coordinates already present in the original polyline representation. No interpolation or artificial points are introduced. Distances are computed after projecting the geographical coordinates into the Belgian Lambert 72 coordinate system (EPSG:31370), allowing accurate metric measurements.

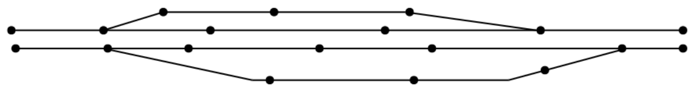

Figure 4.4: Track segmentation Initial setup

For each generated sub-segment, new extremity nodes are created when necessary and the corresponding track attributes are recomputed. The resulting infrastructure therefore preserves the original network geometry while providing a finer representation of the railway infrastructure.

Figure 4.5: Track segmentation applied to a conflicting segment

After segmentation, the platform matching procedure can be repeated on the newly generated segments. The process continues until each station platform is associated with a distinct segment. In practice, a single segmentation iteration was sufficient to eliminate all observed conflicts in the Belgian railway network dataset.

Figure 4.6: Track segmentation results

This approach allows the model to maintain a realistic infrastructure representation while ensuring that stations and platforms can be uniquely mapped to simulation entities. Such a requirement is particularly important for subsequent routing, timetable generation, and railway traffic simulation tasks.

Dual Graph Construction

A dual graph is constructed to support route computation at the micro level. In this representation, each track segment becomes a node of the graph, and each feasible transition between two consecutive track segments becomes a directed edge.

This representation makes it possible to compute routes between platforms rather than only between stations. Dijkstra's shortest-path algorithm is then applied to find the shortest valid path through the dual graph. The resulting sequence of track segments is converted into SUMO edge identifiers and inserted into route definitions.

A route cache is used to avoid recomputing identical paths. Each computed route is stored according to its origin station, destination station, and platform pair. This reduces the computational cost when several trains use the same station-to-station movement.

The construction of the dual graph is performed in two stages.First, each track segment is associated with its start and end junctions. A temporary lookup structure is generated in order to quickly retrieve all tracks connected to a given infrastructure node. This preprocessing step accelerates the graph construction by avoiding repeated searches

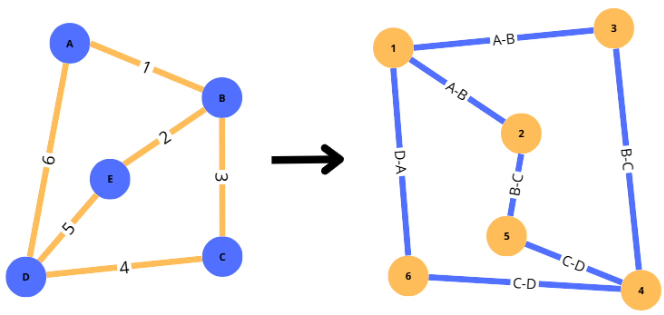

Figure 4.7: Dual graph

through the complete infrastructure dataset.Next, the directed graph is created using the `NetworkX` library. For every pair of track segments, a directed edge is added whenever the destination node of the first track corresponds to the origin node of the second track. Formally, given two track segments (t_i) and (t_j), a directed edge is created if

$$\mathrm{end}(t_i) = \mathrm{start}(t_j). \tag{4.1}$$

The resulting edge represents a valid train movement from one track segment to the next. Each edge is assigned a weight corresponding to the physical length of the originating track segment. Consequently, shortest-path computations minimize the travelled distance across the railway infrastructure.

The resulting graph can be formally defined as $G = (V, E)$,where each vertex $(v \in V)$ represents a railway track segment and each directed edge $((u, v) \in E)$ denotes a feasible transition between two consecutive track segments.

Finally, whenever the operational dataset contains trips between station pairs that have not yet been processed, the corresponding routes are computed on demand and added to the cache. This strategy guarantees that every station-to-station movement required by the timetable is associated with a valid infrastructure path before the simulation is generated.

The dual graph is therefore a central component of the micro-level model. It provides the link between detailed infrastructure reconstruction and operational route generation. However, it also becomes a critical point of failure: if the graph is incomplete or incorrectly connected, real train journeys cannot be mapped onto the reconstructed infrastructure.

4.2.3 Trip Generation

Trip generation defines the train movements executed in the micro-level simulation. Two strategies are considered: random trip generation and real-data-based trip generation.

Random Trips

The random-trip approach generates synthetic train movements on the reconstructed railway network. Its objective is not to reproduce a real timetable, but to test whether the generated infrastructure is connected and usable in SUMO.

Candidate departure and arrival locations are selected from valid station platforms. Weighting files are generated so that SUMO's `randomTrips.py` selects only railway platforms as origins and destinations. Intermediate tracks, switches, and non-operational sections are excluded from the selection process. Additional parameters, such as insertion rate, minimum trip distance, and simulation duration, are used to control the density and realism of the generated traffic.

After trip generation, `duarouter` is used to compute executable routes through the SUMO network. This approach provides a practical way to validate the micro-level infrastructure and verify whether trains can circulate through the reconstructed network.

Real Data Trips

The real-data trip generation strategy aims to reproduce train movements from the punctuality dataset. The operational records are first cleaned and grouped by train identifier and operating day. For each train, station records are sorted according to the planned timetable in order to reconstruct an ordered sequence of stops.

The main difficulty is that each movement between two consecutive stations must be mapped onto a valid path in the detailed infrastructure. For this reason, each station must first be associated with one or more detected platforms. Then, for every pair of consecutive stations, the dual graph is used to compute a path between the selected origin and destination platforms.

Conceptually, this approach is closer to real railway operations than random trip generation because it preserves the structure of actual train services. However, in the current implementation, it did not produce a complete exploitable simulation. Many consecutive station pairs could not be connected because no valid path was found in the generated dual graph. As a result, majority of real train services had to be rejected before route generation.

This issue is an important methodological finding. It shows that high infrastructure coverage is not sufficient to guarantee simulation feasibility. At the micro level, the network must also be topologically consistent and routable. Missing connections, incorrect orientations, or imperfect platform assignments can prevent a route from being reconstructed even when the corresponding movement exists in the real railway network.

4.2.4 Simulation

Once the micro-level infrastructure and trips have been generated, the scenario can be executed in SUMO. The simulation uses detailed railway edges, train stops, route files, vehicle definitions, and configuration files.

Two simulation scenarios are considered. The first one uses random trips. This scenario is mainly used to test the generated network under controlled conditions and to verify that trains can be inserted, routed, and simulated on the reconstructed infrastructure. It provides a useful validation step because it tests the physical connectivity of the generated network without depending on the completeness of real operational routing.

The second scenario attempts to use real-data trips reconstructed from the punctuality dataset. This scenario is more realistic because it aims to reproduce observed railway operations. However, it is also more demanding because every train service must be mapped onto valid platform-to-platform paths in the dual graph. In the current implementation, this requirement could not always be satisfied, preventing the generation of a complete real-data-based micro-level simulation.

The random-trip simulation therefore demonstrates that the generated infrastructure can support train movement in SUMO, while the real-data simulation highlights the remaining difficulty of aligning real operational trajectories with a detailed reconstructed infrastructure.

4.2.5 Limitations

The micro-level model provides a more detailed representation of the railway infrastructure, but it is also more sensitive to data quality and reconstruction errors. Missing connections, incorrect track orientations, imperfect switch detection, or inaccurate platform assignments can prevent valid routes from being computed. These issues are less visible at the macro level, but they become critical when trains must follow continuous physical paths through the network.

The main limitation concerns the integration of real operational data. Random trips can be generated and simulated more easily because they only require feasible routes between selected platforms. Real-data trips are more constrained because each consecutive station pair in the timetable must correspond to a valid path in the dual graph. In the current implementation, this condition was not always satisfied, which prevented the construction of a fully exploitable real-data micro-level simulation.

This limitation does not invalidate the micro-level approach. On the contrary, it reveals an important requirement for railway digital twin construction: a detailed infrastructure model must be evaluated not only by the number of detected elements, but also by its ability to support operational routing. The micro-level model therefore provides both a more realistic infrastructure representation and a clear identification of the challenges that must be solved before real-data-based detailed railway simulation can be achieved.

Chapter 5

Results

This chapter presents the results obtained from the modeling and simulation pipeline developed in this thesis. The evaluation focuses on two main aspects. First, the generated datasets are compared with the available reference data in order to assess their completeness. Second, the infrastructure elements extracted during the modeling process are analysed to evaluate the quality of the generated railway representation.

The results are presented separately for the macro-level and micro-level approaches when necessary. This distinction is important because the two models do not represent the railway network at the same level of detail.

5.1 Dataset Validation

5.1.1 Station Coverage

Station coverage evaluates the number of stations detected and integrated into the generated datasets. This metric is important because stations are central elements in both modeling approaches. In the macro-level model, stations are the main nodes of the graph. In the micro-level model, stations are used to associate platforms and tracks with operational stopping points.

Simulation Level	Generated Stations	Reference Stations	Coverage (%)
Macro-Level	559	559	100
Micro-Level	553	554	99.82

Table 5.1: Station coverage for macro-level and micro-level simulations

The results show that station coverage is very high for both modeling approaches. For the macro-level simulation, all 559 stations present in the reference dataset are generated, corresponding to a coverage of 100.00%. This indicates that, at the station-to-station level, the macro-level generation process is able to fully preserve the set of stations required for the generated network.

For the micro-level simulation, 553 stations are generated from a reference dataset containing 554 stations, corresponding to a coverage of 99.82%. This result is also very strong, especially considering that the micro-level model requires a more detailed association between stations, platforms, and physical track infrastructure. The small difference between the reference and generated datasets may be explained by data inconsistencies, preprocessing filters, or the absence of a valid platform-track association for one station.

Overall, the station coverage results confirm that the generated datasets provide a reliable station basis for the rest of the modeling pipeline. The macro-level approach achieves complete station coverage, while the micro-level approach remains almost complete despite the additional infrastructure constraints introduced by the detailed model.

5.1.2 Platform and Track Coverage

The second part of the dataset validation concerns platform and track coverage. These elements are especially important for the micro-level model, where the objective is to represent the railway infrastructure in more detail. Platforms define where trains can stop inside stations, while tracks define the physical railway segments used for circulation.

Category	Reference Data	Simulation Data	Coverage (%)
Platforms	1551 platforms	1560 platforms	99.42
Tracks (Macro-Level)	1385 tracks	1560 tracks	100
Tracks (Micro-Level)	25028 tracks	26721 tracks	106.59

Table 5.2: Platform and track coverage for micro-level simulation

The platform coverage obtained for the micro-level model is also very high. The reference dataset contains 1560 platforms, while the generated dataset contains 1551 platforms, resulting in a coverage of 99.42%. This shows that the platform detection and matching process is able to identify almost all platforms available in the reference data. However, it does not guarantee that the generated platforms are correctly positioned on the tracks, which is a separate aspect of the infrastructure modeling process.

Track coverage differs between the macro-level and micro-level models. In the macro-level simulation, the reference data contains 1385 station-to-station tracks and the generated dataset also contains 1385 tracks, giving a coverage of 100%. This result is expected because the macro-level model directly represents connections between stations and does not attempt to reconstruct the detailed internal railway topology.

For the micro-level simulation, the generated dataset contains 26721 tracks, while the reference dataset contains 25068 tracks. This corresponds to a coverage of 106.59%. A value higher than 100% does not necessarily indicate an error. It reflects the fact that the micro-level generation process may create more detailed or segmented track representations than those present in the reference dataset. In particular, a single reference track can be represented by multiple generated segments after preprocessing, switch detection, and track segmentation.

Overall, the platform and track coverage results show that the generated infrastructure is highly complete. However, the interpretation of track coverage must take into account the difference between reference tracks and simulation-ready SUMO edges. The generated micro-level network is not only a copy of the reference data, but a transformed representation adapted to routing and simulation constraints.

5.2 Infrastructure Modeling Results

After validating the general completeness of the datasets, the next step is to evaluate the results of the infrastructure modeling process. This part focuses on the railway elements that were extracted, transformed, or reconstructed during the implementation. The objective is to assess whether the generated infrastructure provides a coherent basis for railway simulation.

The infrastructure modeling results confirm the high level of coverage already observed in the dataset validation step. Stations are almost completely detected in both modeling approaches, and platforms are also recovered with a high success rate in the micro-level model. Switches require a more nuanced interpretation, because they are inferred from the generated railway topology rather than directly copied from the reference dataset.

Category	Reference Data	Simulation Data	Detection (%)
Stations (Macro-Level)	559	559	100
Stations (Micro-Level)	554	553	99.82
Platforms (Micro-Level)	1560	1551	99.42
Switches (Micro-Level)	10379	12343	118.92

Table 5.3: Detected stations, platforms and switches for macro-level and micro-level simulations

The detection percentage reaches 100.00% for macro-level stations and 99.82% for micro-level stations. Platform detection reaches 99.42%, which confirms that most platforms could be matched to the generated infrastructure. These results are important because platforms are used as stopping locations for trains and as origin or destination points during trip generation.

Switch detection provides a different type of result. The reference data contains 10379 switches, while the generated micro-level infrastructure contains 12343 detected switches, corresponding to a coverage of 118.92%. As with micro-level track coverage, this value above 100% must be interpreted carefully. It does not simply mean that the model detects more correct switches than expected. Instead, it indicates that the algorithm identifies more candidate switch points than the number of switches available in the reference dataset.

This visual comparison makes it possible to understand how the algorithm enriches the infrastructure by identifying branching and merging points across the network.

This difference can be explained by the way switches are inferred from the railway topology. In the generated network, a switch is detected when a node connects more than two track segments and therefore behaves as a branching or merging point. However, this topological definition may identify some elements that are not classified as switches in the reference data. It may also split complex junctions into several smaller switch-like elements.

The visual comparison of the network before and after switch detection helps illustrate this effect. The switch detection process enriches the generated infrastructure by identifying relevant branching points and making the network more suitable for routing. In dense areas such as Bruxelles-Midi, the detected switches provide a much more detailed representation of the railway topology, but they also highlight the difficulty of matching generated topological elements exactly with reference infrastructure objects.

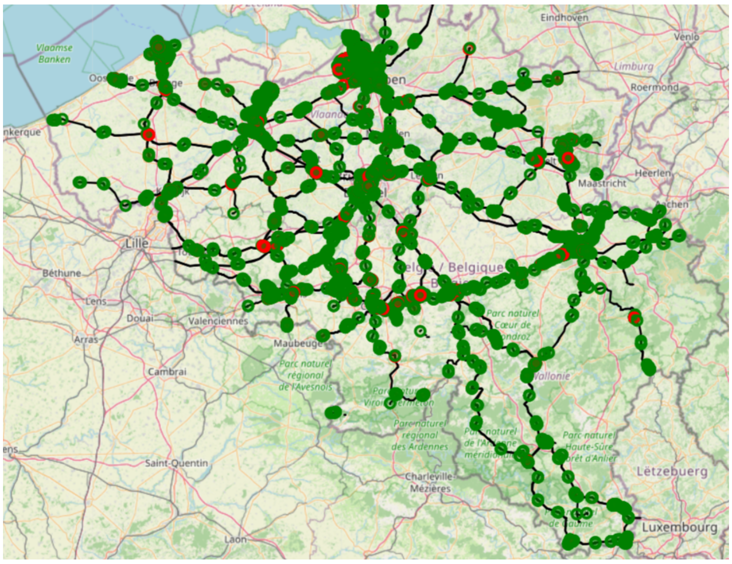

Figure 5.1: Network before and after switch detection (Green dots represent detected switches with 3 connections and Red dots represent detected switches with 4 or more connections)

This station is particularly relevant because it contains a dense set of tracks and platforms, making it a useful example for evaluating whether the platform matching algorithm can handle complex station environments.

The platform matching figures further illustrate the quality of the generated infrastructure. They show how platforms are associated with the corresponding railway tracks and make it possible to visually inspect the quality of the matching process. This qualitative validation complements the numerical coverage metrics and helps identify local cases where the matching may be less accurate.

The platform matching quality is further illustrated by a qualitative comparison of several representative stations. As shown in Figure 5.4, the matching process can produce different levels of quality depending on the local infrastructure configuration. For instance, Tubize is classified as a good-quality match, with a very small minimum distance between the detected platform and the candidate track, which suggests a reliable association. Weerde corresponds to an intermediate case, labelled as a warning, where the matching remains acceptable but presents a larger average distance, indicating a less precise association. By contrast, Hansbeke is classified as insufficient quality, showing that the algorithm may encounter difficulties in some local configurations, especially when the search radius is larger or when the number of candidate tracks does not allow a clear correspondence to be established.

5.3 Simulation Results

The previous sections evaluated the completeness of the generated datasets and the quality of the reconstructed infrastructure. This section focuses on the simulation results obtained from the two implemented modeling approaches. As in the implementation chapter, the results are separated between the macro-level and micro-level simulations because the two models do not have the same objective or the same level of detail.

The macro-level simulation is evaluated through a real-data scenario executed on a

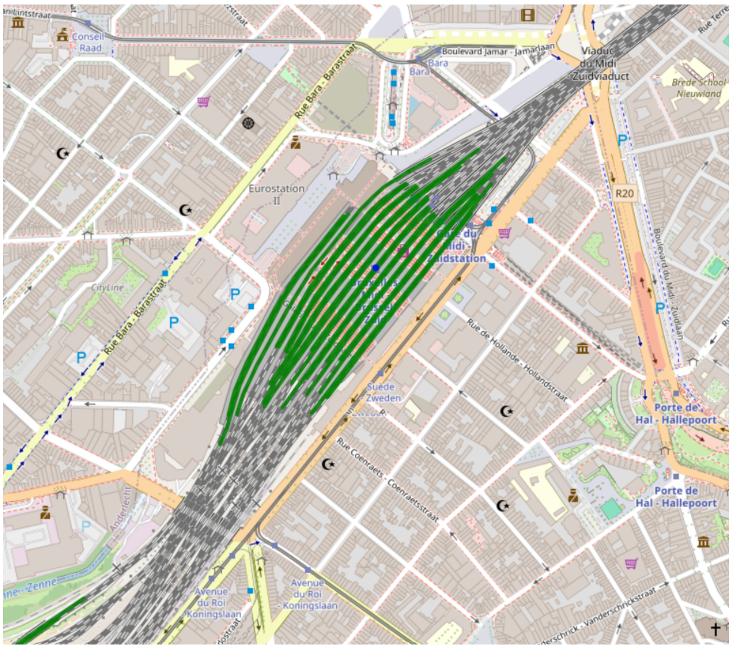

Figure 5.2: Platform matching result on Bruxelles-Midi

small chain of neighbouring stations. The micro-level simulation is evaluated through a random-trip scenario executed on the reconstructed detailed network. These two results illustrate the current simulation capabilities of the proposed framework and highlight the different roles of the two levels of representation.

5.3.1 Macro-Level Simulation Results

The macro-level simulation was successfully executed on a selected chain of four neighbouring stations: Cour-sur-Heure, Ham-sur-Heure, Berzée, and Pry. This scenario was chosen because it corresponds to a limited and coherent part of the railway network, with moderate traffic intensity and a simple station-to-station structure.

The simulation uses real operational data extracted from the punctuality dataset. Train movements are reconstructed from the observed sequence of stations and converted into SUMO routes. This means that the simulated traffic is not randomly generated: it corresponds to real train services that circulated through the selected part of the network during the considered time window. The result therefore validates the ability of the macro-level pipeline to connect infrastructure data, operational data, route generation, and SUMO simulation execution.

Element	Value
Simulated Area	Cour-sur-Heure, Ham-sur-Heure, Berzée, Pry
Number of Stations	4
Trip generation method	Real operational data
Simulation duration	5 days
Number of generated train routes	115
Number of complete simulated trips	115

Table 5.4: Macro-level simulation scenario summary

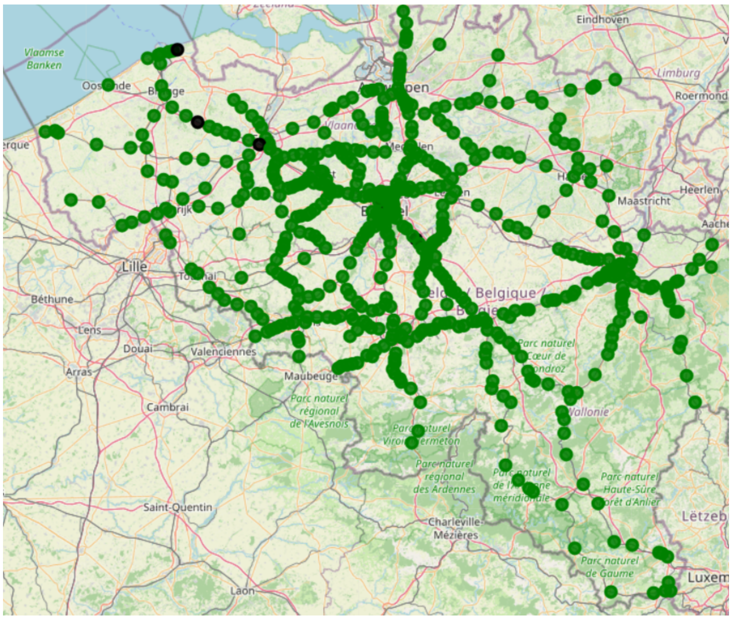

Figure 5.3: Platform matching quality map

Figure 5.4: Platform matching quality examples

The main result of this scenario is that the macro-level approach is able to execute a simulation based on real train movements, provided that the selected part of the network remains compatible with the assumptions of the model. In particular, the approach works best on simple chains of neighbouring stations where each station has a limited local complexity. Stations with two main directions are especially suitable because the simplified station-node representation remains close to the actual operational structure. In contrast, dense stations with multiple platforms, branching directions, and high traffic intensity are less appropriate for this abstraction because internal routing and platform constraints are not represented.

This result confirms the usefulness of the macro-level model as a first validation layer for the proposed DTR. It demonstrates that real operational data can be transformed into executable SUMO routes when the selected network area is sufficiently simple and coherent. However, the result should not be interpreted as a complete validation of real railway operations. The macro-level simulation validates station-to-station circulation, but it does not validate detailed infrastructure behaviour inside stations.

5.3.2 Micro-Level Simulation Results

The micro-level simulation was evaluated using randomly generated trips over the reconstructed detailed railway network. As explained in Chapter 4, the real-data micro-level scenario could not be fully executed because the mapping of real train movements onto the

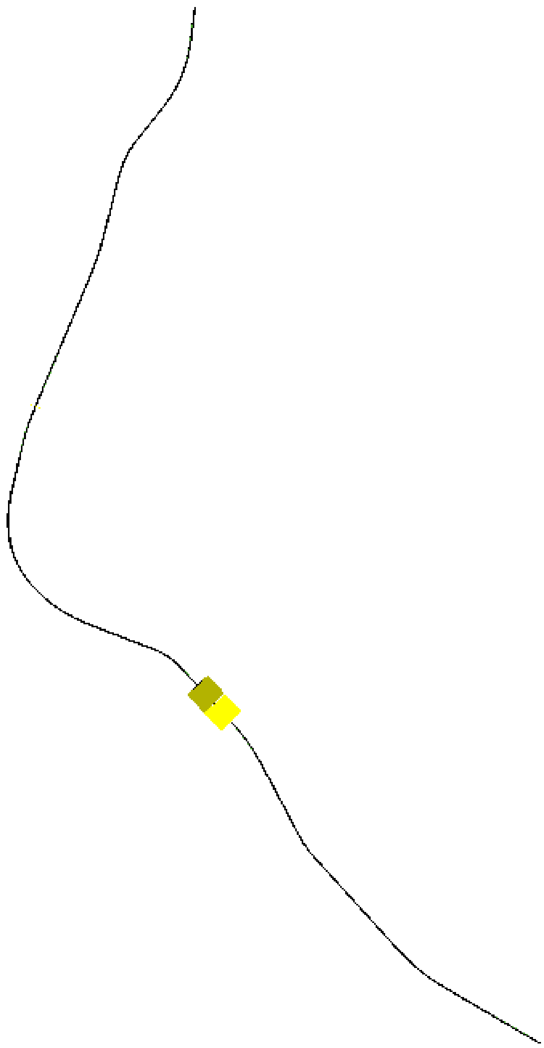

Figure 5.5: Macro-level simulation network for the selected four-station chain

detailed infrastructure graph did not provide valid paths for a sufficient number of station pairs. For this reason, the result presented here focuses on the random-trip scenario.

The random-trip simulation was executed on the entire reconstructed micro-level network. Trips were generated between valid platform locations and then routed through the SUMO network. The objective of this scenario is therefore not to reproduce a real timetable, but to evaluate whether the generated infrastructure can support train circulation at the micro level.

The results show that 16802 train routes were generated and successfully routed. This is an important result because it confirms that the generated infrastructure is not only a static representation of railway tracks, but can also be used by SUMO to compute executable train movements. In this sense, the random-trip simulation validates the connectivity of the generated network and demonstrates that the reconstructed micro-level infrastructure is usable for simulation purposes.

However, the number of completed trips is significantly lower than the number of generated and routed trips. Out of 16802 generated routes, 4912 trips were completed during the simulation. This difference must be interpreted carefully. It does not necessarily indicate that the generated network is disconnected, since the routes were successfully

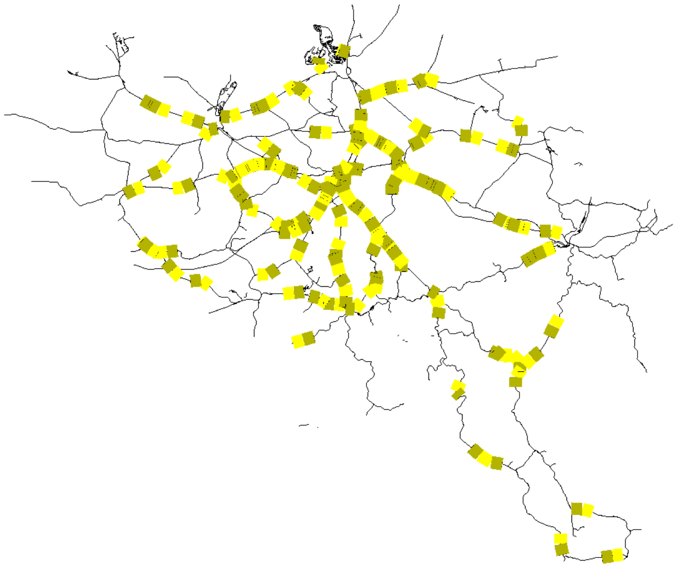

Figure 5.6: SUMO-GUI screenshot of the micro-level random trip simulation

Element	Value
Simulated Area	Entire reconstructed micro-level network
Number of Stations	553
Number of Platforms	1551
Number of Tracks	26721
Trip generation method	Random trips between valid platforms
Simulation duration	1 day
Number of generated train routes	16802
Number of successfully routed trips	16802
Number of completed trips	4912

Table 5.5: Micro-level simulation scenario summary

computed. Rather, it reflects the limitations of random traffic generation when applied to a detailed railway network.

Random trips are generated without reproducing the full set of operational constraints that would normally be considered in a real railway timetable. In particular, the generation process does not fully account for train speeds, headways, platform occupation, route conflicts, junction capacity, or the temporal separation between incompatible movements. As a result, many trains may be inserted too close to one another or may attempt to use conflicting parts of the infrastructure at similar times. During the simulation, this can lead to warnings such as emergency braking, collisions, teleportation events, or blocked insertion areas.

```
Warning: Block after insertion lane '29646_0' exceeds maximum length
(stopped searching after edge ':2794_0' (length=20207.42m)).
Warning: Vehicle '12897' performs emergency braking on lane '49375_0'
with decel=5.00, wished=1.00, severity=1.00, time=85488.00.
Warning: Teleporting vehicle '12897'; collision with vehicle '10232',
lane='49527_0', gap=-40.28, time=85489.00, stage=move.
```

These warnings are therefore consistent with the nature of the generated scenario. They show that the infrastructure can be used for routing, but that random trip generation does not provide a realistic operational timetable. The simulation should consequently be interpreted as a technical validation of the generated network rather than as a realistic reproduction of railway operations. It demonstrates that trains can circulate on the reconstructed micro-level infrastructure, while also showing that a more constrained and timetable-aware trip generation method is required to obtain operationally meaningful results.

Overall, the micro-level simulation result confirms the feasibility of executing train movements on the detailed network. The fact that all generated trips could be routed suggests that the network contains a high level of connectivity from the point of view of SUMO routing. However, the lower number of completed trips highlights that routing feasibility alone is not sufficient to produce realistic railway simulation. Future improvements should therefore focus not only on infrastructure connectivity, but also on the generation of traffic scenarios that better account for railway operational constraints.

5.4 Computational Performance

In addition to evaluating the quality of the generated railway infrastructure and the execution of simulation scenarios, it is also important to analyse the computational performance of the proposed pipeline. This evaluation provides insight into the cost of transforming the initial railway datasets into simulation-ready SUMO configuration files. It also helps identify which steps of the implementation are the most time-consuming and where future optimizations should be prioritized.

The performance analysis is divided into two complementary parts. The first part focuses on the generation pipeline, from the initial ETL operations to the creation of the network, route, stop, and configuration files required by SUMO. The second part focuses on the execution of the simulation itself, with particular attention to the micro-level model, which contains a more detailed infrastructure representation and is therefore more computationally demanding.

5.4.1 Pipeline Generation Performance

The first evaluation concerns the execution time of the main processing steps required to generate a simulation scenario. For both the macro-level and micro-level models, the pipeline can be decomposed into several stages: data loading and preprocessing, infrastructure generation, route or trip generation, additional file generation, and final simulation configuration.

The longer execution time of the micro-level pipeline is therefore expected. It is mainly caused by the larger number of infrastructure elements and by the additional processing stages required to construct a simulation-ready detailed network. In particular, the micro-level pipeline includes switch detection, platform detection, track segmentation, detailed network generation, and random trip generation. These operations are not required, or are much simpler, in the macro-level model.

The comparison between the macro-level and micro-level generation pipelines shows a clear difference in computational cost. The macro-level pipeline completes in approximately 331 seconds, while the micro-level pipeline requires approximately 775 seconds.

Pipeline Step	Execution Time (s)
ETL	330.92
Network generation	0.12
Trip generation	15.08
Additional file generation	0.0058
Configuration generation	0.0000
Total	331.0458

Table 5.6: Execution time of the macro-level generation pipeline for the example network

Pipeline Step	Execution Time (s)
ETL	344.09
Switch detection	1.50
Platform Detection	19.07
Track segementation	81.90
Network generation	160.37
Trip generation	168.30
Additional file generation	0.02
Configuration generation	0.00
Total	775.25

Table 5.7: Execution time of the micro-level generation pipeline with random trip generation

This result is coherent with the structure of the two models. The macro-level model manipulates a simplified station-to-station graph, whereas the micro-level model processes a much larger and more detailed infrastructure representation.

Among the micro-level processing steps, network generation and trip generation are particularly costly. This is consistent with the fact that the micro-level network contains many more track segments and requires more complex route computation than the macro-level model. Track segmentation also introduces a non-negligible cost, since it modifies the structure of the infrastructure in order to increase the operational granularity of the network.

This result highlights the trade-off between abstraction and realism. The macro-level model is faster and more robust for real-data simulation on selected station chains, but it cannot represent detailed infrastructure constraints. The micro-level model is more expensive to generate, but it provides a richer infrastructure representation that can support more detailed simulations. The additional computational cost is therefore coherent with the higher level of detail introduced by the micro-level approach.

5.5 Discussion

The results presented in this chapter show that the proposed pipeline is able to generate and simulate railway networks at two complementary levels of representation. The macro-level model provides a simplified but robust framework for integrating real operational data into SUMO. The micro-level model provides a more detailed representation of the physical railway infrastructure and demonstrates that train circulation can be simulated on a network composed of tracks, switches, platforms, and segmented railway sections.

The dataset validation results indicate that the generated infrastructure is highly complete. Station coverage is almost complete for both modeling levels, and platform coverage is also very high. Track coverage must be interpreted more carefully, especially for the micro-level model, where the number of generated tracks exceeds the number of reference tracks. This does not necessarily indicate an error. It reflects the fact that the generated SUMO network is a transformed and more segmented representation of the original infrastructure data. The objective is not only to copy the reference dataset, but to create a simulation-ready representation that can be used for routing and train movement simulation.

The infrastructure modeling results also show the value of the micro-level approach. Switch detection and platform matching make it possible to enrich the raw infrastructure data and transform it into operational simulation artefacts. The platform matching quality examples show that the matching process can produce different levels of reliability depending on the local configuration of the station. Some stations provide clear and accurate matches, while others require larger search radii or present more ambiguous candidate tracks. This confirms that the method is effective overall, but that local infrastructure complexity remains an important factor.

The simulation results highlight the complementarity between the two modeling levels. At the macro level, real operational data can be used directly on selected station chains, provided that the selected area remains compatible with the simplifying assumptions of the model. This confirms the usefulness of the macro-level approach as a first end-to-end validation of the digital twin workflow. It demonstrates that infrastructure data, punctuality data, route generation, simulation execution, and output processing can be connected in a coherent pipeline.

At the micro level, the results are different but equally informative. The random-trip simulation confirms that the detailed network can be used by SUMO to route and execute train movements. The fact that 16802 trips were generated and successfully routed suggests that the reconstructed infrastructure is connected from a routing perspective. However, only 4912 trips were completed during the simulation. This difference shows that network connectivity alone is not sufficient to guarantee a realistic and stable railway simulation.

The lower number of completed trips is mainly explained by the nature of random trip generation. Random trips are useful for testing network connectivity and simulation feasibility, but they do not reproduce the constraints of a real timetable. They do not fully account for train speeds, headways, route conflicts, junction capacity, platform occupation, or safe temporal separation between trains. As a consequence, the simulation may generate operational conflicts, leading to emergency braking, teleportation events, insertion problems, or collision-related warnings. These events should not be interpreted as a failure of the infrastructure generation alone, but as evidence that random traffic generation is not sufficient to reproduce realistic railway operations.

The computational performance results further support the distinction between the two approaches. The macro-level pipeline is faster because it manipulates fewer entities and relies on a simpler graph structure. The micro-level pipeline requires more time because it processes a larger amount of data and performs additional operations such as switch detection, platform matching, track segmentation, network generation, and route generation. This additional cost is coherent with the increased realism of the model. It reflects a natural trade-off: the macro-level model is more scalable and easier to use with real operational data, while the micro-level model provides a richer infrastructure basis but requires more processing and more careful traffic generation.

These findings also clarify the next challenge for the proposed digital twin. The micro-level network appears sufficiently connected to support random routing, but the real-data routing strategy based on the dual graph did not yet provide exploitable timetable-based simulation results. This suggests that the next difficulty is not only to generate a connected network, but to identify the correct physical paths corresponding to real train movements. The dual graph approach provides a useful methodological foundation, but future work may need to explore alternative routing strategies, improved direction handling, better platform assignment, or precomputed station-to-station path libraries.

Overall, the results show that the proposed framework successfully establishes the foundations of a railway digital twin for simulation. The macro-level model validates the integration of real operational data in a simplified setting, while the micro-level model validates the reconstruction and simulation of a detailed infrastructure network. The current implementation does not yet provide a fully realistic micro-level timetable simulation, but it identifies the main technical obstacles that must be addressed. In this sense, the contribution of the thesis is not limited to the final simulation outputs, it also lies in the structured analysis of the trade-offs between abstraction, realism, computational cost, and operational feasibility.

Chapter 6

Limitations

This chapter discusses the main limitations encountered during the development of the proposed railway digital twin simulation pipeline. Some limitations were already briefly introduced in the implementation chapter, but they are analysed here in more detail. The objective is not only to identify what prevented the system from reaching a complete operational simulation, but also to clarify which issues are related to data availability, modeling assumptions, routing strategies, simulator constraints, and generalization.

The proposed framework successfully demonstrates that railway infrastructure and operational data can be transformed into SUMO-compatible simulation artefacts. However, the results also show that building a realistic railway simulation from heterogeneous real-world datasets remains a difficult task. This is especially visible at the micro level, where the objective is to reproduce the detailed physical structure of the network and to map real train movements onto this infrastructure. The limitations discussed in this chapter therefore provide important insights for future work on railway digital twins.

6.1 Data Availability and Quality

A first important limitation concerns the availability and precision of the infrastructure data. Although the available datasets provide a strong basis for reconstructing the railway network, some information required for a fully realistic simulation is either missing, incomplete, or not available at the level of detail required by the simulator. This issue is particularly important for the micro-level model, where the simulation depends on the accurate representation of tracks, switches, platforms, and their spatial relationships.

One of the clearest examples concerns platform positions. The available data provides information about stations and the number of platforms, but it does not always provide the exact physical position of each platform in the railway infrastructure. As a result, platforms had to be reconstructed using geometric methods, mainly by associating each station platform with nearby track segments. This approach makes it possible to generate platform candidates and to create SUMO train stops, but it also introduces uncertainty. Even when the correct number of platforms is detected, their exact placement may not correspond to the real infrastructure. A platform may be associated with a neighbouring track instead of the operationally correct one, especially in complex stations where several tracks are close to each other.

This limitation has a direct impact on simulation quality. In SUMO, train stops must be attached to specific lanes. If a platform is associated with an incorrect lane, the train may stop at a location that is geographically close to the station but operationally unrealistic. In some cases, the selected track may also be difficult or impossible to reach from the reconstructed route. Therefore, platform detection is not only a data enrichment step, but also a critical dependency for the feasibility of the micro-level simulation.

Another limitation concerns operational route data. The punctuality dataset describes train passages through stations and provides temporal information such as planned and observed arrival or departure times. However, it does not provide the exact physical route followed by each train inside the railway infrastructure. In other words, the data indicates that a train moved from one station to another, but it does not specify which tracks, switches, platforms, or junctions were actually used. This absence of route-level information is one of the main obstacles to constructing a complete micro-level simulation from real data.

Because of this limitation, train paths had to be reconstructed manually or algorithmically from station sequences. This task is considerably more complex at the micro level than at the macro level. Without reference data describing the possible or historical paths used by trains, the reconstruction process must infer these paths from infrastructure topology alone. This makes the result highly dependent on the completeness and correctness of the generated graph.

A dataset describing possible train paths, authorized routes, or historical physical trajectories would significantly improve the pipeline. Such a dataset would provide a direct link between operational records and infrastructure elements. It would also make it possible to distinguish between several possible routes connecting the same pair of stations, which is essential in dense railway areas. Without this information, the simulation can reconstruct plausible paths, but it cannot guarantee that these paths correspond to the routes actually used in railway operations.

6.2 Modeling Assumptions

The proposed methodology relies on two levels of abstraction: a macro-level model and a micro-level model. This distinction is useful because it makes it possible to compare a simple and robust representation with a more detailed but more fragile one. However, each level introduces specific modeling assumptions that limit the realism of the resulting simulation.

At the macro level, stations are represented as single nodes and railway links as direct edges between stations. This abstraction is useful for integrating real operational data, because punctuality records are also expressed mainly at the station level. However, this representation hides the internal structure of stations. Platforms, switches, parallel tracks, junctions, signals, and local routing constraints are not represented explicitly. Consequently, the macro-level model cannot reproduce conflicts occurring inside stations, platform occupation constraints, or detailed route interactions near complex junctions.

The macro-level model is therefore mainly suitable for selected station chains with moderate traffic. It provides a functional end-to-end validation of the pipeline, but it cannot be interpreted as a complete operational simulation of the railway system. This is why the macro-level simulation was restricted to selected parts of the network instead of attempting to reproduce the full national traffic. In dense areas, the internal structure of stations and junctions becomes too important to be ignored.

At the micro level, the opposite limitation appears. The model represents the infrastructure in greater detail, but this increases the sensitivity of the pipeline to small data errors. Missing connections, incorrect directions, ambiguous platform assignments, or imperfect switch detection can prevent valid routes from being found. In a detailed infrastructure graph, a small topological inconsistency may invalidate an entire train journey, because the route generation process requires a continuous path between each pair of consecutive stations.

Another important modeling assumption concerns the use of geometric proximity for platform matching. The method assumes that the closest relevant track segments are likely to correspond to the station platforms. This assumption is often reasonable, but it does not always hold in complex railway environments. Large stations, yards, parallel tracks, underground sections, or curved station layouts can create ambiguous spatial configurations. In such cases, geometric proximity alone may not be sufficient to identify the correct operational platform.

These assumptions show that the micro-level model should be understood as a simulation-ready reconstruction of the railway infrastructure rather than as an exact copy of the real network. The objective was to generate a usable infrastructure model from available data, but the absence of some operational and infrastructure details limits the degree of realism that can be achieved.

6.3 Route Reconstruction and Micro-Level Simulation

Another group of limitations is related to the use of SUMO as the simulation platform. SUMO provides a flexible and scriptable environment for generating networks, defining routes, running simulations, and extracting outputs. This flexibility was one of the main reasons for selecting it in this thesis. However, SUMO was not originally designed as a railway-specific operational planning tool. Some railway-specific concepts therefore require additional modeling effort.

In particular, detailed signalling, interlocking, dispatching, platform assignment, route reservation, and railway capacity management are not automatically reproduced unless they are explicitly encoded in the generated files or controlled through additional logic. This is an important limitation for the micro-level simulation, because detailed infrastructure alone is not sufficient to reproduce realistic railway operations. A network may be physically connected, but the simulation may still generate unrealistic behaviour if operational constraints are not represented.

This issue is visible in the random-trip simulation. The generated trips can be successfully routed through the network, but they are not based on a real timetable and do not fully account for headways, platform occupation, train speeds, junction capacity, or safe temporal separation between incompatible movements. As a consequence, trains may be inserted too close to one another or may attempt to use conflicting infrastructure sections at similar times. This can lead to warnings such as emergency braking, blocked insertion areas, teleportation events, or collision-related messages.

These warnings should not be interpreted only as failures of the generated infrastructure. They also reflect the limitations of unconstrained random trip generation in a detailed railway network. In real railway operations, train movements are planned and regulated to avoid such conflicts. Without an equivalent timetable generation or dispatching layer, the simulator receives routes and departure times that may be technically valid but operationally unrealistic.

A further limitation concerns calibration. The simulation parameters used in SUMO, such as speed, dwell time, insertion rate, routing parameters, and trip generation settings, influence the behaviour of the simulation. In the current work, these parameters were selected to obtain plausible and executable scenarios, but they were not calibrated against a complete operational reference model. As a result, the simulation can be used to validate the feasibility of the pipeline, but not yet to provide precise quantitative estimates of railway performance.

6.4 Simulation Constraints in SUMO

The proposed pipeline also faces limitations related to computational complexity. The macro-level approach is relatively efficient because it manipulates a reduced graph composed mainly of stations and station-to-station links. This makes it easier to generate routes from real operational data and to execute simulations on selected station chains.

The micro-level approach is more demanding. It processes a much larger number of infrastructure elements and applies additional operations such as switch detection, platform matching, track segmentation, network generation, and route computation. This additional cost is expected because the model provides a richer representation of the infrastructure. However, it also limits the ease with which the approach can be scaled to large scenarios with real operational traffic.

The difference between routing feasibility and simulation completion is also important. A route can be computed successfully, but the corresponding train may still fail to complete its trip during simulation because of timing conflicts, blocked lanes, unrealistic insertion times, or interactions with other trains. This means that scalability is not only a question of graph size or computation time. It also depends on the quality of the generated timetable and on the operational consistency of the simulated traffic.

For this reason, future versions of the pipeline would need to include additional mechanisms for timetable validation, route conflict detection, and traffic regulation before simulation execution. These mechanisms would help ensure that the generated scenario is not only connected, but also operationally feasible.

6.5 Scalability and Computational Complexity

The final limitation concerns the generalization of the proposed approach. The pipeline was developed and evaluated using the available Belgian railway datasets and the specific structure of the data used in this thesis. Although the general methodology could be transferred to other railway networks, several components would require adaptation.

First, infrastructure datasets may differ significantly between countries or infrastructure managers. Station identifiers, track geometries, platform descriptions, switch representations, and coordinate systems may follow different conventions. The preprocessing and matching rules developed in this work may therefore not be directly applicable to another network.

Second, operational data may also differ in structure and granularity. Some datasets may contain only planned timetables, while others may include real-time train positions, platform assignments, delay causes, or detailed route information. The quality of the resulting digital twin would depend strongly on the type of operational data available.

Third, the assumptions used in the macro-level and micro-level models may not be equally valid for all railway systems. A simple station-to-station abstraction may be sufficient for low-density regional lines, but less appropriate for dense metropolitan networks. Similarly, a micro-level reconstruction based on geometric proximity may work well in some areas but fail in complex multi-level stations or yards.

The proposed framework should therefore be seen as a methodological foundation rather than a universal solution. It demonstrates how railway infrastructure and operational data can be transformed into simulation-ready artefacts, but it also shows that the quality of the final simulation depends on the availability of detailed data, the correctness of the reconstructed infrastructure, the realism of the routing strategy, and the operational constraints represented in the simulator.

Overall, these limitations do not invalidate the contribution of the thesis. On the contrary, they clarify the main technical barriers that must be addressed to move from a simulation-oriented prototype toward a more complete railway digital twin. The macro-level model validates the integration of real operational data in a simplified setting, while the micro-level model identifies the additional requirements needed for realistic infrastructure-based simulation. Future work should therefore focus on improving platform localization, integrating route-level operational data, replacing single shortest-path routing with multiple feasible path strategies, and adding timetable and conflict-management mechanisms to the simulation pipeline.

Chapter 7

Conclusion

This thesis addressed the construction of a simulation-oriented digital twin for railway network simulation using SUMO and heterogeneous railway datasets. The objective was not to develop a complete operational digital twin, but rather to study the foundations required to transform infrastructure and operational data into simulation-ready railway models.

The work focused on two complementary levels of representation. The macro-level model represents stations as nodes and railway connections as edges, while the micro-level model represents infrastructure elements such as tracks, platforms, switches, and operational segments. This distinction made it possible to evaluate the trade-off between abstraction, scalability, realism, and simulation feasibility.

7.1 Summary of Contributions

The first contribution of this thesis is the design and implementation of a complete data-driven pipeline for railway simulation generation. This pipeline includes data preprocessing, network generation, trip generation, simulation execution, and output analysis. It provides a structured methodology for converting railway datasets into SUMO-compatible simulation artefacts.

The second contribution is the development of a macro-level railway model. This model provides a simplified but functional representation of the railway network and allows real operational data to be integrated into simulation scenarios. It validates the complete workflow from punctuality data processing to SUMO simulation.

The third contribution is the development of a micro-level railway model. This model reconstructs the railway infrastructure in greater detail by representing tracks, switches, platforms, and segmented track sections. It also introduces routing structures, such as the dual graph, that support path computation inside the detailed infrastructure.

Finally, this thesis contributes by identifying the main technical challenges involved in railway digital twin construction. These challenges include data quality, infrastructure completeness, platform matching, route reconstruction, SUMO-specific constraints, and computational scalability.

7.2 Key Findings

The results show that a railway simulation-oriented digital twin can be constructed progressively by combining different levels of abstraction. The macro-level model proved to be more robust for integrating real operational data, since train movements can be represented as station-to-station trajectories. This makes it suitable for validating the overall pipeline and for running simulations on selected parts of the network.

The micro-level model provides a more realistic infrastructure representation, but it also introduces greater complexity. Although the generated infrastructure reaches high coverage and can support synthetic train movements, real-data-based simulation remains difficult. The main issue is that real train trajectories require valid and continuous paths between all consecutive stations, which is not always guaranteed in the reconstructed micro-level graph.

The results also highlight that high dataset coverage does not automatically ensure operational correctness. A station, platform, or track may be detected, but still be insufficiently aligned with the routing requirements of a real railway service. Therefore, railway digital twin construction requires not only good data coverage, but also topological consistency and operational validity.

Overall, the main finding is that macro-level and micro-level models are complementary. The macro-level model is better suited for robust and scalable simulation based on real train movements, while the micro-level model provides a foundation for more detailed infrastructure analysis.

7.3 Practical Implications

This work shows that existing railway data can be transformed into executable simulation models through a structured processing pipeline. This is useful for researchers and railway stakeholders who want to analyse railway operations without manually constructing simulation networks.

The macro-level model can serve as a practical tool for preliminary traffic analysis, selected line simulation, and validation of operational data integration. Its simplified structure makes it easier to use and more robust when detailed infrastructure data is not required.

The micro-level model, although still limited for real-data simulation, provides a basis for more detailed studies of railway infrastructure. With further improvements, it could support analyses related to platforms, switches, local bottlenecks, routing conflicts, and infrastructure capacity.

The work also shows that SUMO can be used as a flexible platform for railway digital twin prototyping. However, railway-specific elements such as signaling, interlocking, dispatching, and precise platform assignment require additional modeling beyond the standard SUMO workflow.

7.4 Future directions

The work presented in this thesis provides a first implementation of a railway digital twin based on real infrastructure and operational data. However, several extensions could be considered in order to transform the current framework into a more complete decision-support tool.

7.4.1 Counterfactual Simulation and What-If Scenarios

A first direction concerns counterfactual simulation and what-if scenarios. The current framework mainly reproduces existing infrastructure and train movements. A future extension could allow users to modify parts of the system and observe the effect of these changes on railway traffic.

For example, it could be possible to simulate the closure of a station, the removal of a track section, a modification of train frequency, or a change in departure times. At the macro level, this could be implemented by modifying station-to-station connections. At the micro level, it could be implemented by making specific platforms, switches, or track segments unavailable.

This would make the digital twin useful as an exploratory tool for testing operational decisions before applying them to the real network.

7.4.2 Tracks Works Planning

A second direction concerns track works planning. Maintenance operations often require temporary closures, reduced capacity, speed restrictions, or rerouting. A simulation-based digital twin could help evaluate the impact of these interventions before they take place.

In the current framework, this could be implemented by adding availability constraints to infrastructure elements. At the macro level, railway connections could be removed or penalized. At the micro level, specific tracks, platforms, or switches could be marked as unavailable during a given time window.

This extension would help compare maintenance strategies, identify bottlenecks, and reduce the impact of planned works on train operations.

7.4.3 Real-Time Forecasting and Incident Propagation

A third direction is the integration of real-time forecasting and incident propagation. The current implementation is mainly based on historical and static datasets. A future version could integrate live or near-real-time information, such as train delays or incidents, and use it to predict the future state of the network.

This could be implemented by combining the simulation pipeline with statistical or machine learning models trained on punctuality data. The macro-level model could be used to study delay propagation between stations, while the micro-level model could help analyse the role of local infrastructure constraints.

Such an extension would make the digital twin useful for operational monitoring and proactive traffic management.

7.4.4 Framework Generalization

A final direction concerns the generalization of the framework. The current implementation is adapted to the Belgian railway datasets used in this thesis. A more general version should separate the core modeling logic from dataset-specific processing steps.

This could be achieved by introducing modular data adapters that transform different railway datasets into a common internal representation. Depending on the available data, the framework could then generate a macro-level model, a micro-level model, or a hybrid representation.

This would increase the reproducibility of the approach and make it easier to apply the methodology to other railway networks.

7.5 Final Remarks

This thesis has shown that constructing a railway digital twin simulation is an incremental process. Before advanced capabilities such as real-time prediction or decision support can be achieved, it is necessary to build reliable simulation models from available infrastructure and operational data.

The proposed macro-level and micro-level approaches demonstrate two complementary ways of addressing this challenge. The macro-level model provides a robust and scalable representation for integrating real train movements, while the micro-level model provides a more detailed infrastructure representation and highlights the challenges of realistic route reconstruction.

The main conclusion of this work is that railway digital twin development requires a balance between simplicity and realism. A simplified model is easier to simulate and more robust, but cannot represent detailed operational constraints. A detailed model is more realistic, but depends more strongly on data completeness and topological correctness.

Although the current implementation is not yet a complete operational railway digital twin, it provides a solid foundation for future work. It demonstrates that railway data can be transformed into SUMO-compatible simulation artefacts, that real-data macro-level simulation is feasible, and that micro-level infrastructure reconstruction can support synthetic railway traffic. Future improvements should focus on route reconstruction, infrastructure consistency, railway-specific operational constraints, and the integration of predictive or real-time data.

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